



CSE471: System Analysis and Design
Project Report
Project Title: Project Title Here

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Submission Date:

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1. System Request

Business need:

Current urban management systems often struggle with delayed communication and a lack of accountability. Citizens frequently find it difficult to report local issues like infrastructure damage or waste mismanagement, while authorities lack a centralized tool to prioritize these tasks effectively. This project addresses the need for a transparent, automated, and real-time platform to bridge the gap between residents and city officials.

Business requirements:

The system must provide a secure portal for verified citizens to report issues with location data and evidence. It needs to automate the classification of these reports using natural language processing and manage them through a structured resolution workflow. Key features include automated Service Level Agreement (SLA) tracking, emergency broadcasting, and data visualization through heatmaps for city planners.

Business value:

- **Increased Efficiency:** Reduces manual data entry and routing through automated classification.
- **Enhanced Accountability:** Tracks the performance of specific departments and provides public progress updates.
- **Improved Safety:** Enables rapid response through geo-targeted emergency alerts and resource tracking.
- **Data Driven Insights:** Helps authorities identify high incident zones to better allocate city budgets and resources.

Special issues or constraints:

- **Identity Verification:** Users must provide valid NID, Passport, or Birth Certificate details to post, ensuring high quality data and accountability.
- **Technical Dependencies:** Relies on third party services like Google Maps for location tracking and various APIs for translation and NLP.
- **Data Sensitivity:** Must securely handle personal identification information and geographic data.

2. Functional Requirements

Module 1: Complaint Submission and Processing

1. **Submission Engine:** Allows verified users to submit multimedia complaints with GPS tagging, unique ticket IDs, and a 4 minute edit window.
2. **AI Classification:** Uses NLP to automatically detect issue types, extract keywords, and suggest the responsible department.
3. **Integrity Control:** Implements automated spam filtering and duplicate detection based on geographic proximity and keyword analysis.
4. **Multilingual Support:** Provides a bilingual interface (Bangla/English) with real-time translation for all complaint descriptions.

Module 2: Intelligent Priority, SLA and Validation

1. **Smart Prioritization:** Automatically sets priority levels (Low to Critical) and visualizes issue intensity via Google Maps heatmaps.
2. **Notification Hub:** Sends real-time alerts (SMS/Email/In-app) and broadcasts geo-targeted emergency notifications for high-risk issues.
3. **SLA Monitoring:** Assigns resolution deadlines based on category and tracks remaining time for each active ticket.
4. **Automated Escalation:** Moves overdue tasks to higher authorities while utilizing AI to filter fake or prank submissions.

Module 3: Monitoring, Transparency and Smart Intelligence

1. **Emergency and Resource Management:** Enables geo-targeted disaster broadcasts and dynamic resource allocation suggestions based on demand.
2. **Citizen Portal:** Provides a public dashboard for real-time status tracking, department history, and aggregated civic analytics.
3. **Departmental Workspace:** Offers a central dashboard for officials to update tickets, add field notes, and provide proof of resolution.
4. **Workflow Engine:** Manages and logs state transitions (Pending to Closed) through a configurable process tracker.
5. **Performance Analytics:** Generates reports on resolution times, department efficiency, and monthly SLA compliance statistics.
6. **Civic Feedback Loop:** Features a rating system for resolution quality and a reputation-based badge system for active users.
7. **Executive Dashboard:** Provides the Mayor a high-level view of city performance, heatmap analysis, and a volunteer recruitment system.
8. **Reporting and Audit:** Generates comprehensive PDF reports including specialized corruption risk predictions based on case delays and reassignments.

3. Technology (Framework, Languages)

- **Language:** JavaScript
- **Frameworks:** MERN Stack (MongoDB, Express.js, React.js, Node.js)

4. Backend Development

Attach at least 5 code snippets of Backend/API development with proper description. Also, attach corresponding screenshots of Postman API testing.

(i) This API endpoint performs an atomic update on the User model after validating the phone number against Bangladeshi regex standards. It ensures data integrity by

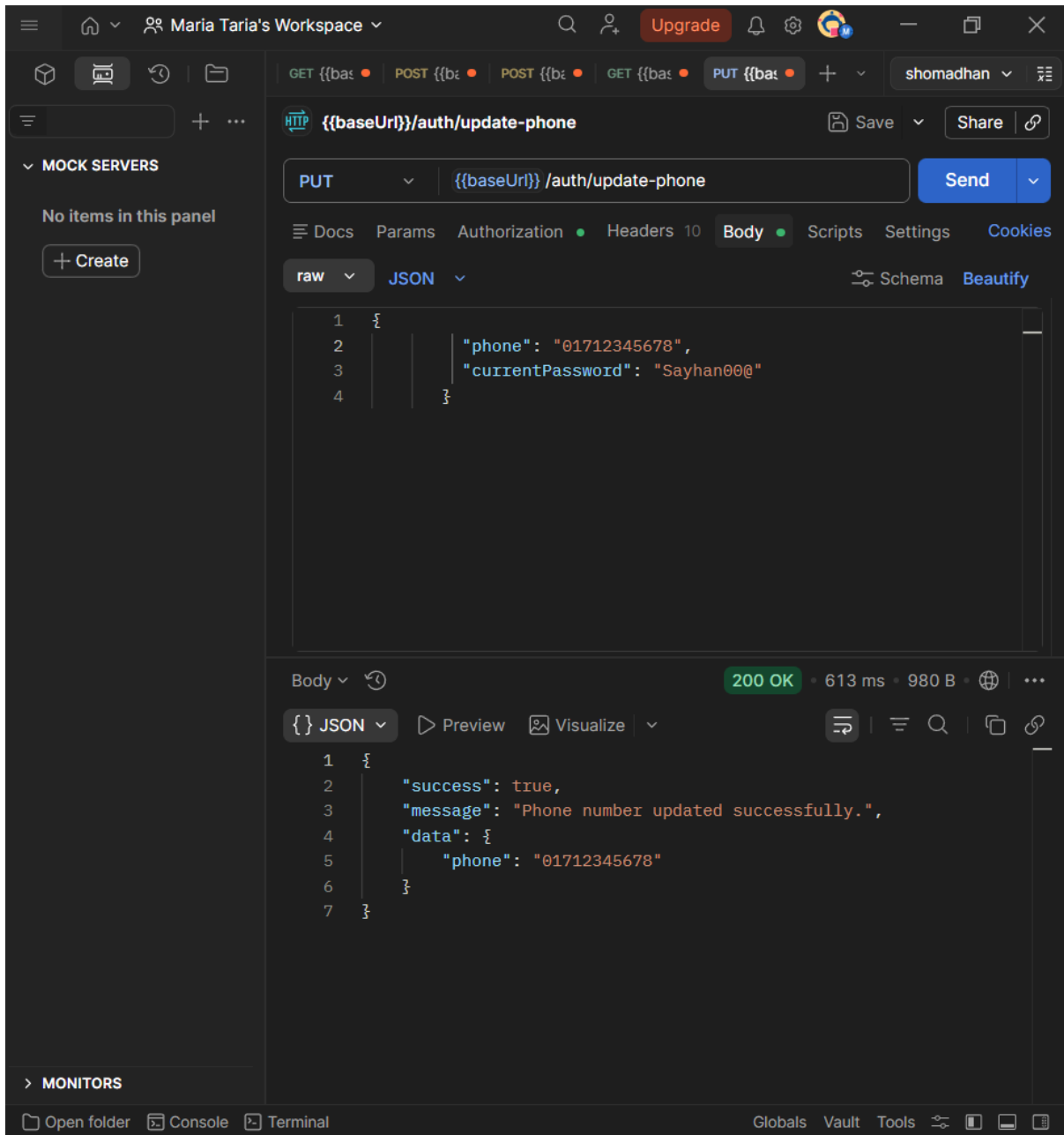
restricting modification rights to the authenticated user ID extracted from the JWT session

```
// PUT /api/v1/auth/update-phone
exports.updatePhone = async (req, res, next) => {
  try {
    const { phone } = req.body;

    // Validating BD phone format before saving
    if (!/^(\\+880|0)?1[3-9]\\d{8}$/.test(phone)) {
      return res.status(400).json({ success: false,
message: "Invalid phone number" });
    }

    const user = await User.findByIdAndUpdate(
      req.user.id,
      { phone },
      { new: true, runValidators: true }
    );

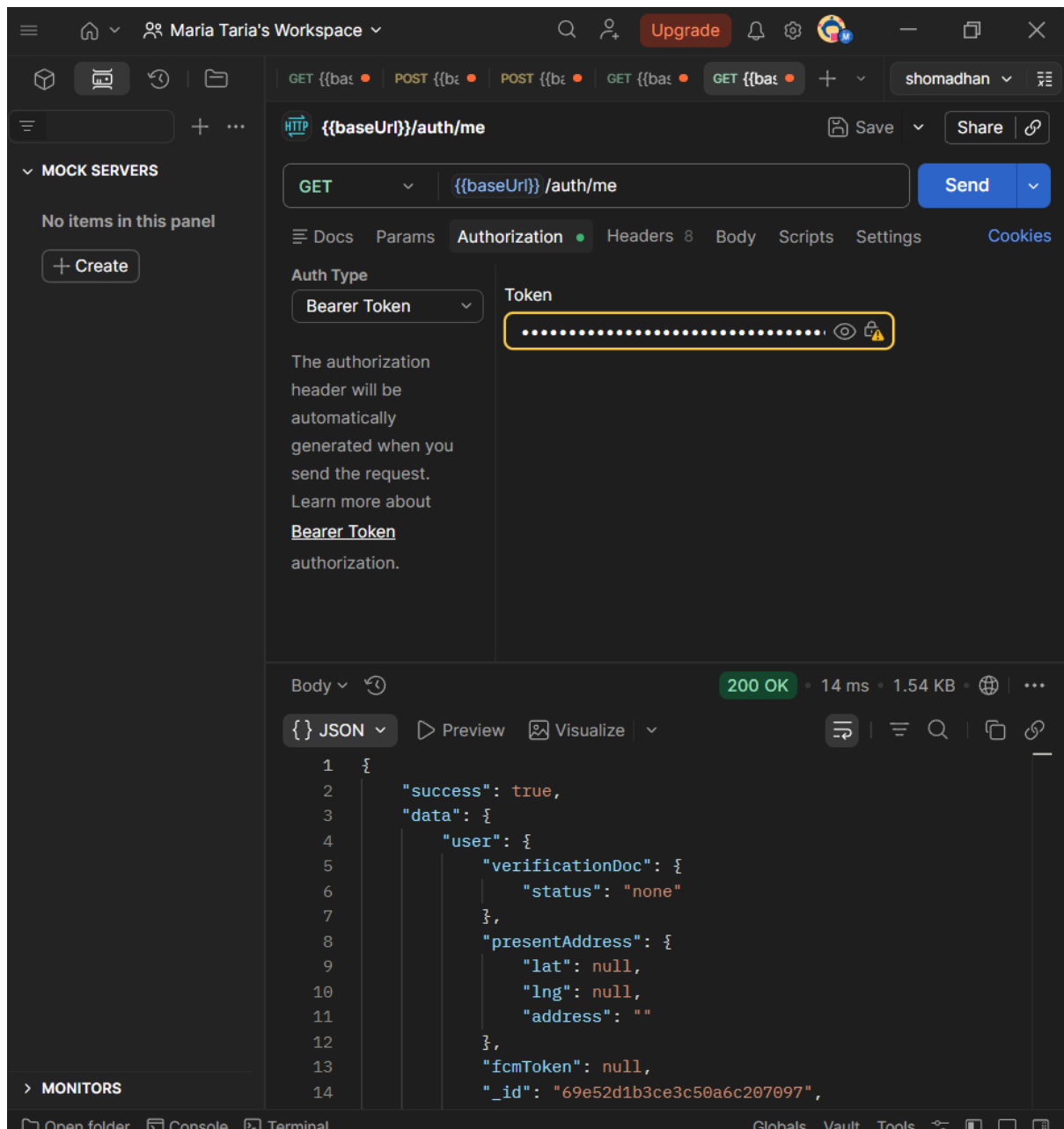
    res.status(200).json({ success: true, data: user });
  } catch (err) {
    next(err);
  }
};
```



(ii) A critical security endpoint that retrieves the full state of the currently authenticated user from the database. It serves as the primary data source for the frontend to determine user permissions, roles, and reputation-based features.

```
// GET /api/v1/auth/me
exports.getMe = async (req, res, next) => {
  try {
    // req.user.id is populated by the protect
    middleware
    const user = await User.findById(req.user.id);

    res.status(200).json({
      success: true,
      data: user, // Returns name, role, reputation, and
      badges
    });
  } catch (err) {
    next(err);
  }
};
```



(iii) This backend service utilizes the PDFKit library to programmatically generate and stream binary PDF data directly to the client. It aggregates real-time data from the Complaint collection to produce a visual accountability heatmap for city officials.

```

// PDF Logic for Case Briefing & Risk Audit
exports.generateDepartmentReport = async (req, res,
next) => {
  const doc = new PDFDocument({ size: 'A4', layout:
'landscape', margin: 40 });

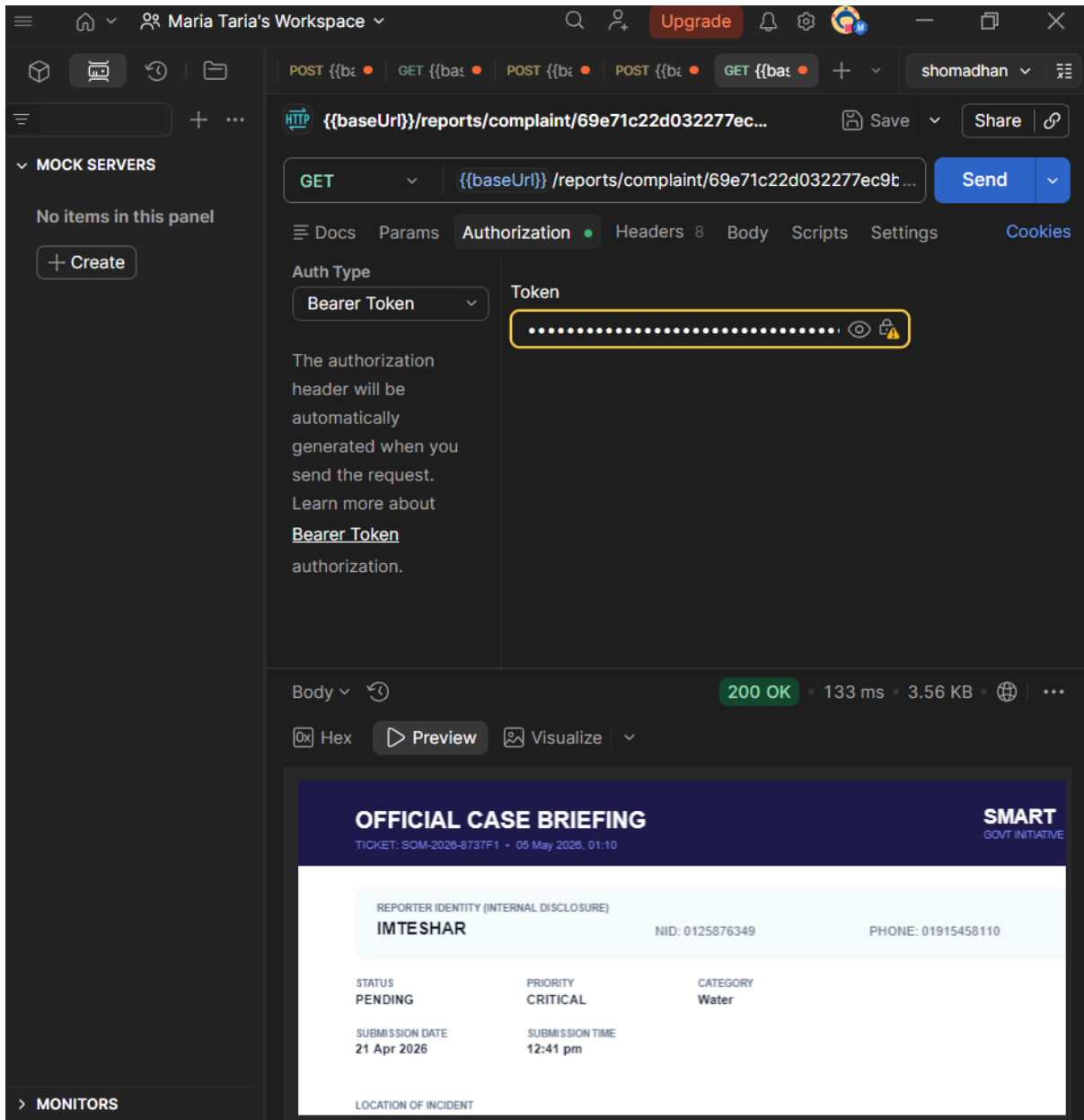
  // Header Design
  doc.fillColor('#0f172a').font('Helvetica-
Bold').fontSize(22).text('SHOMADHAN AUDIT REPORT', 40,
50);
  doc.fontSize(10).font('Helvetica').text(`Generated on:
${new Date().toLocaleString()}`, 40, 80);

  // Accountability Heatmap Logic
  const riskData = await calculateRiskScores(); //
Custom algorithm
  riskData.forEach((r, i) => {
    const rowColor = r.level === 'High' ? '#fef2f2' :
'#f0fdf4';
    doc.roundedRect(40, 150 + (i * 40), 762, 35,
6).fill(rowColor);

    doc.fillColor('#1e293b').text(`${r.key.toUpperCase()} -
${r.score}% RISK`, 55, 162 + (i * 40));
  });

  doc.pipe(res); // Streaming PDF directly to
browser/Postman
  doc.end();
};

```



(iv) A complex multi-part API that processes binary file uploads (audio/video/image) while simultaneously calling an internal AI service for department classification. It persists the evidence metadata and AI-assigned categories into the MongoDB database.

```

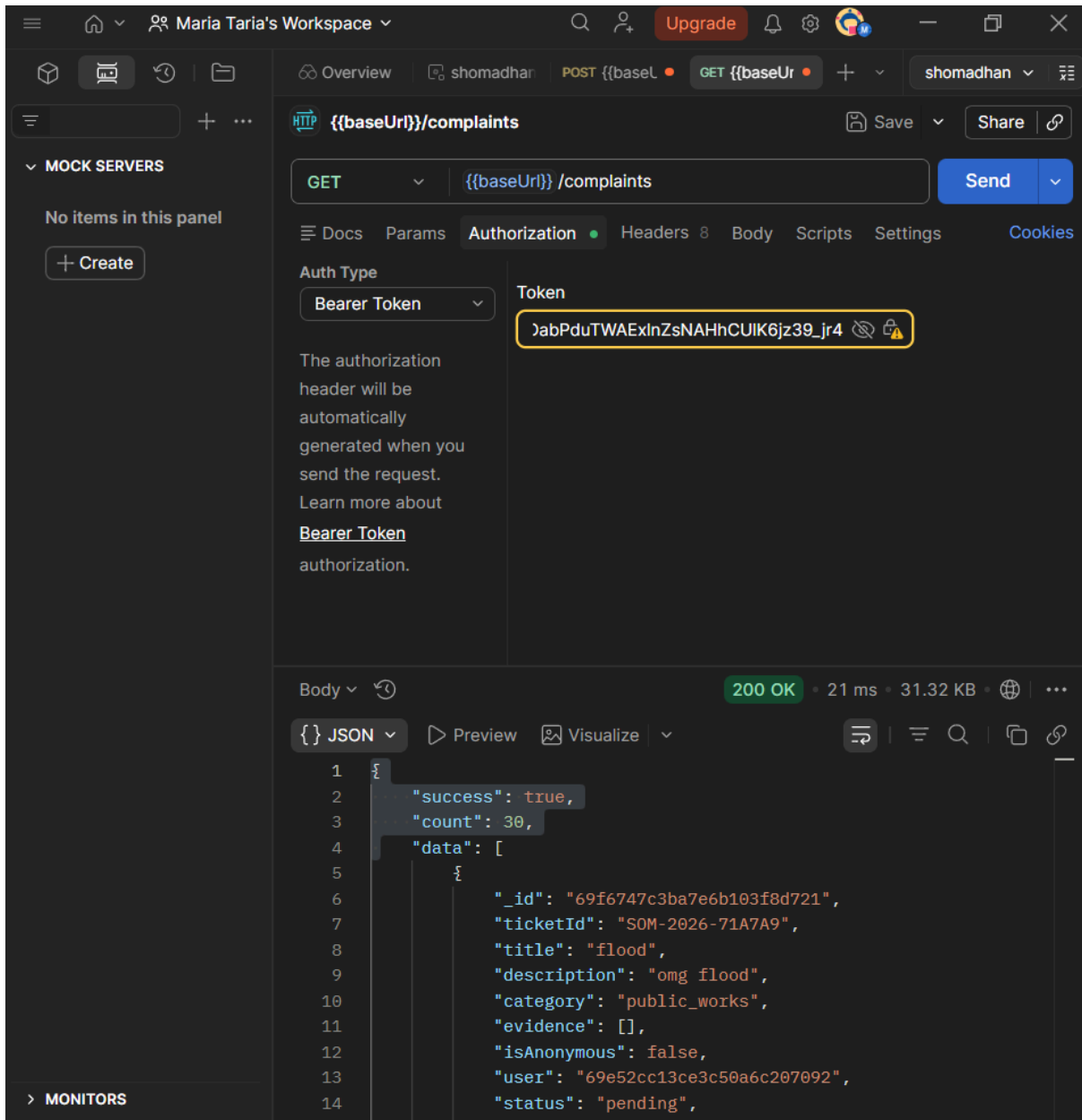
// POST /api/v1/complaints
exports.createComplaint = async (req, res, next) => {
  // Mapping uploaded file paths (Voice, Image, Video)
  const evidence = req.files ? req.files.map(f =>
`/uploads/${f.filename}`) : [];

  // AI-Driven Department Classification
  const analysis = await
classifyComplaint(req.body.title, req.body.description);

  const complaint = await Complaint.create({
    ...req.body,
    user: req.user.id,
    evidence,
    category: analysis.department.key, // AI Assigned
    aiConfidence: analysis.confidence
  });

  res.status(201).json({ success: true, data: complaint
});
};

```

(v) The primary authentication entry point that performs encrypted password verification using bcrypt. It implements account-state checks (active/inactive) and issues a signed JSON Web Token (JWT) to establish a secure, stateless user session.

```

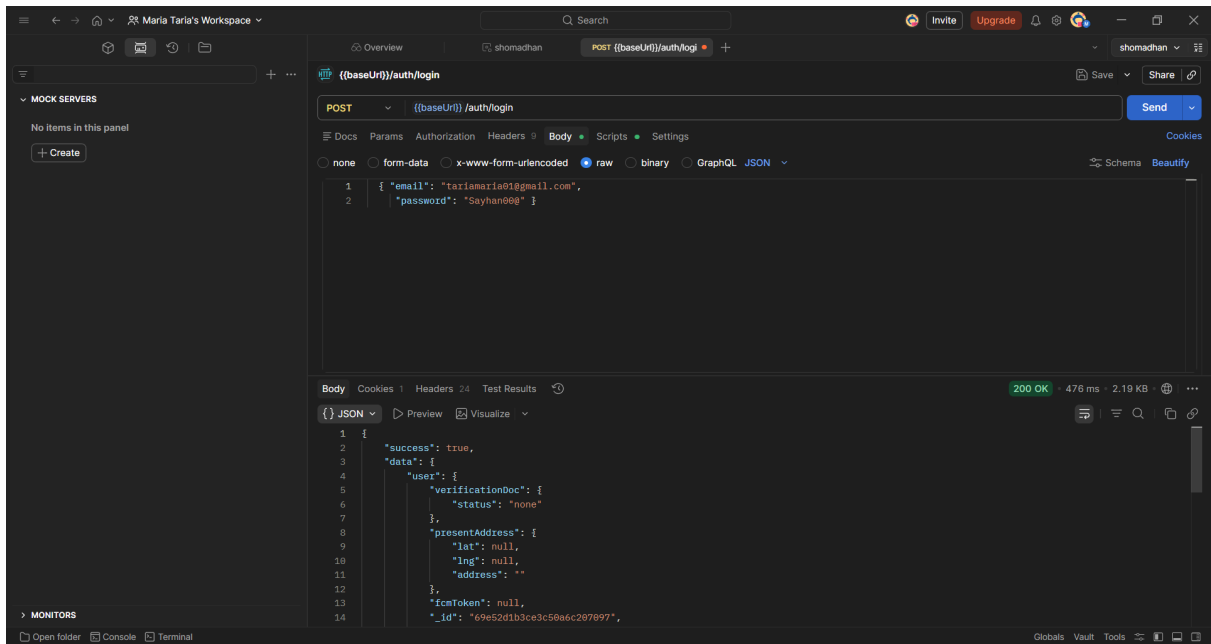
// POST /api/v1/auth/login
exports.login = async (req, res, next) => {
  const { email, password } = req.body;

  // 1. Check if user exists and include the hashed
  password
  const user = await User.findOne({ email
}).select('+password');
  if (!user || !(await user.comparePassword(password)))
  {
    return res.status(401).json({ success: false,
message: "Invalid credentials" });
  }

  // 2. Check if account is approved/active
  if (!user.isActive) {
    return res.status(403).json({ success: false,
message: "Account pending approval" });
  }

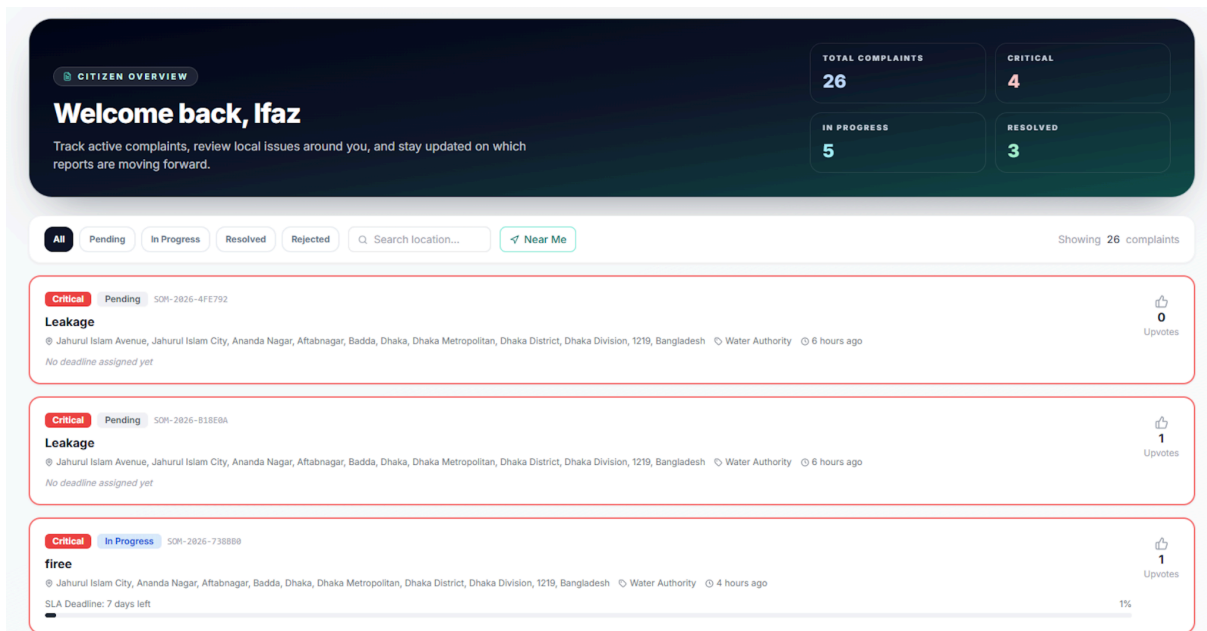
  // 3. Generate and send JWT Token
  const token = user.generateToken();
  res.status(200).json({ success: true, token });
};

```



5. User Interface Design

Attach at least 5 interface designs generated in Figma. Attach the figma project link.



COMMUNITY VOICE

Feedback Dashboard

Review community-submitted feedback on resolved complaints without leaving your dashboard.

INSIGHT FOCUS
See how citizens describe service quality after resolution.

Community Feedback

Total Feedback
1

Overall Rating
4.7

Response Time
5.0

Professionalism
5.0

Q Search ticket, complaint, location, citizen, o

All ratings

1 entries shown

SOM-2026-DEE673 4.7/5 overall

★★★★★

View Complaint

Flood
Imteshar Ahmed water_authority South Badda, Gulshan, Dhaka, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1212, Bangladesh 04 May 2026

Resolution Quality
★★★★☆

Response Time
★★★★★

Officer Professionalism
★★★★★

good

REPORT CENTRE

City Operations

Generate downloadable PDF reports - individual complaint case files or aggregated city-wide summaries with charts, SLA data, and executive metrics.

City Efficiency

Summary Reports

Individual Case Report

Select Complaint

Q Search by ticket ID or title...

SOM-2026-4FE792
Leakage pending

SOM-2026-B18E0A
Leakage pending

SOM-2026-738BB0
fire in-progress

SOM-2026-8EED74
moon spotted pending

SOM-2026-75CEAC

EMERGENCY BROADCAST CONTROL PANEL

Public Servant Emergency Desk

Send geo-targeted disaster alerts to citizens whose saved address falls inside the selected emergency radius.

TARGETING RULE
 Only citizens with saved map coordinates inside the affected radius will receive the alert.

Create Emergency Broadcast
 Fill in the incident details, choose the impact area, and send the alert.

ALERT TITLE

DISASTER TYPE

EMERGENCY MESSAGE

State what happened, the risk, and what citizens should do immediately.

AFFECTED AREA

Locate Area

RADIUS (KM)

EXACT RADIUS

LIVE RADIUS
3.0
 kilometers

Choose the emergency center point

Click anywhere on the map to update the target zone

Activity Center

Notifications
 Track account activity, verification updates, and emergency or complaint alerts in one place.

INBOX SUMMARY
23
 All account notes and live alerts in one feed

UNREAD ALERTS
0
 Updates still waiting for review

LATEST ACTIVITY
May 5
 Most recent database activity

Recent Updates
 0 unread alerts, 2 account notes, and 21 live feed updates.

Logged In Successfully SYSTEM
 You signed in on Tuesday, May 5, 2026 at 12:57 AM.
May 5, 12:57 AM

Complaint Received: SOM-2026-8EED74
 Your complaint "moon spotted" has been received and is currently pending review. Ticket ID: SOM-2026-8EED74
May 5, 12:04 AM

View Details →

Complaint Received: SOM-2026-4CC5F8
 Your complaint "alien in my area" has been received and is currently pending review. Ticket ID: SOM-2026-4CC5F8
May 4, 11:47 PM

View Details →

Complaint Received: SOM-2026-090A1F
 Your complaint "moon spotted" has been received and is currently pending review. Ticket ID: SOM-2026-090A1F
May 4, 11:46 PM

View Details →

Figma link:

<https://www.figma.com/design/YaoVv9MePZU8wRPwSbYuVM/471?node-id=30-1171&t=2N1pYU2RkySSIjP-1>

Frontend Development

Attach at least 5 frontend code snippets with proper description.

(i) Interactive Map Radius Targeting

```

import { MapContainer, TileLayer, Marker, Circle, useMapEvents } from "react-leaflet";
import L from "leaflet";
import "leaflet/dist/leaflet.css";

// 1. Initial State and Configuration
const DHAKA_CENTER = [23.8103, 90.4125];
const [mapPosition, setMapPosition] = useState(DHAKA_CENTER);
const [radiusKm, setRadiusKm] = useState(2); // Default 2km radius

// 2. Custom Map Event Handler: Allows user to click on the map to set location
const MapEvents = () => {
  useMapEvents({
    click(e) {
      const { lat, lng } = e.latlng;
      setMapPosition([lat, lng]);
      // reverseGeocode is called here to fetch area name based on coordinates
      fetchAreaLabel(lat, lng);
    },
  });
  return null;
};

// 3. The Interactive UI Component
return (
  <div className="h-[450px] relative rounded-[2rem] overflow-hidden border border-slate-200 shadow-xl group">
    <MapContainer
      center={mapPosition}
      zoom={13}
      className="h-full w-full z-0"
    >
      <TileLayer
        url="https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png"
        attribution='&copy; <a
href="https://www.openstreetmap.org/copyright">OpenStreetMap</a> contributors'
      />

      { /* Visualizing the Impact Zone with a Semi-Transparent Circle */ }
      <Circle
        center={mapPosition}
        radius={radiusKm * 1000} // Converting KM to Meters
        pathOptions={{
          color: '#f43f5e', // Rose-500
          fillColor: '#f43f5e',
          fillOpacity: 0.2,
          weight: 2,
          dashArray: '5, 10' // Animated/Dashed border effect
        }}
      />
    </MapContainer>
  </div>
);

```

```

    {/* Precision Draggable Marker */}
    <Marker
      position={mapPosition}
      draggable={true}
      icon={defaultIcon}
      eventHandlers={{
        dragend: (e) => {
          const marker = e.target;
          const position = marker.getLatLng();
          setMapPosition([position.lat, position.lng]);
          fetchAreaLabel(position.lat, position.lng);
        }
      }}
    />

    <MapEvents />
  </MapContainer>

  {/* Radius Control Slider Overlay */}
  <div className="absolute bottom-6 left-6 right-6 z-[1000] bg-white/90 backdrop-blur-md
    p-4 rounded-2xl border border-slate-200 shadow-lg">
    <div className="flex items-center justify-between mb-2">
      <span className="text-xs font-black uppercase tracking-widest text-slate-500">
        Targeting Radius: <span className="text-rose-600">{radiusKm} KM</span>
      </span>
    </div>
    <input
      type="range"
      min="0.5"
      max="15"
      step="0.5"
      value={radiusKm}
      onChange={(e) => setRadiusKm(parseFloat(e.target.value))}
      className="w-full h-1.5 bg-slate-200 rounded-lg appearance-none cursor-pointer
        accent-rose-500"
    />
  </div>
</div>
);

```

Description: This snippet demonstrates the integration of React-Leaflet for geospatial alerts. It allows an official to drag a marker or click on a map to define an emergency zone, visually rendering a red circle to represent the broadcast's impact radius in kilometers.

(ii) The Hybrid Voice Engine


```

// Implementation of Hybrid Voice (Speech-to-Text + Raw Audio Recording)
const toggleSpeech = async () => {
  if (isRecording) {
    stopAll(); // Custom cleanup function to stop both STT and Recorder
    return;
  }

  try {
    // 1. Initialize High-Fidelity Audio Capture
    const stream = await navigator.mediaDevices.getUserMedia({ audio: true });
    const recorder = new MediaRecorder(stream);
    audioChunks.current = [];
    recorder.ondataavailable = (e) => {
      if (e.data.size > 0) audioChunks.current.push(e.data);
    };
    recorder.onstop = () => {
      const audioBlob = new Blob(audioChunks.current, { type: 'audio/webm' });
      const audioFile = new File([audioBlob], `voice-note-${Date.now()}.webm`, { type:
'audio/webm' });
      if (onAudioRecorded) onAudioRecorded(audioFile); // Passing audio blob to parent form
      stream.getTracks().forEach(track => track.stop());
    };
    recorder.start();
    mediaRecorderRef.current = recorder;

    // 2. Initialize Speech Recognition (STT)
    const SpeechRecognition = window.SpeechRecognition || window.webkitSpeechRecognition;
    const recognition = new SpeechRecognition();
    recognition.lang = lang; // Supports bn-BD or en-US
    recognition.interimResults = true;
    recognition.continuous = true;

    recognition.onresult = (event) => {
      let sessionTranscript = "";
      for (let i = event.resultIndex; i < event.results.length; ++i) {
        sessionTranscript += event.results[i][0].transcript;
      }
      // Merging the new session transcript with the existing field value
      const finalValue = lastBaseValue.current
        ? `${lastBaseValue.current.trim()} ${sessionTranscript.trim()}`
        : sessionTranscript.trim();
      onChange(finalValue);
    };

    recognition.start();
    recognitionRef.current = recognition;
  } catch (err) {
    console.error("Voice system failed:", err);
    setIsRecording(false);
  }
};

```

Description: This is the complete real-time transcription and audio capture logic. It uses useRef to maintain the Speech Recognition state across re-renders and handles the complex task of appending live speech to existing text without data loss.

(iii) Multi-Step Form Logic & AI Analysis

```
// Step Navigation and Data Consolidation
const nextStep = () => {
  if (step === 1 && (!formData.title || !formData.description)) {
    return toast.error("Please provide a title and description");
  }
  setStep((prev) => prev + 1);
};

// AI Classification Trigger: Analyzing text to find the correct department
const analyzeIncident = async () => {
  if (!formData.title || !formData.description) return;
  setAnalyzing(true);
  try {
    const res = await complaintAPI.analyze(formData.title, formData.description);
    const { department, confidence, topCategories } = res.data.data;

    // Applying AI suggestion to the form and updating UI
    setFormData(prev => ({
      ...prev,
      department: department.key,
      category: topCategories[0].category
    }));
    toast.success(`Department Detected: ${department.name} (${Math.round(confidence * 100)}% Match)`);
  } catch (error) {
    toast.error("AI could not determine department automatically.");
  } finally {
    setAnalyzing(false);
  }
};
```

Description: This snippet shows the state management for a 3-step complaint submission process, including the logic that triggers the backend AI for automated department classification.

(iv) Role-Based Navigation & Live Notification Badge System

```
const ServantSidebar = () => {
  const { user, logout } = useAuth();
  const [unreadCount, setUnreadCount] = useState(0);
  const [isCollapsed, setIsCollapsed] = useState(false);

  // NavItem configuration with dynamic Badges
  const navItems = [
    { path: "/servant/dashboard", label: "Dashboard", icon: LayoutDashboard },
    {
      path: "/servant/complaints",
      label: "My Department",
      icon: ClipboardList,
      // Dynamic badge showing pending work items
      badge: unreadCount > 0 ? unreadCount : null
    },
    { path: "/servant/emergency", label: "Emergency Control", icon: AlertTriangle }
  ];

  return (
    <motion.nav
      animate={{ width: isCollapsed ? 80 : 288 }}
      className="h-screen bg-white border-r flex flex-col shadow-2xl transition-all"
    >
      {navItems.map((item) => {
        const active = location.pathname === item.path;
        return (
          <Link key={item.path} to={item.path} className={`nav-link ${active ? 'active' : ''}`>
            <item.icon size={20} className={active ? 'text-blue-600' : 'text-slate-500'} />

            {/* Animate Text Labels for Collapsible UI */}
            {!isCollapsed && (
              <motion.span initial={{ opacity: 0, x: -10 }} animate={{ opacity: 1, x: 0 }}>
                {item.label}
              </motion.span>
            )}

            {/* Live Count Badge for Urgent Tasks */}
            {!isCollapsed && item.badge && (
              <span className="ml-auto bg-rose-500 text-white text-[10px] font-bold px-2 py-0.5 rounded-full animate-pulse">
                {item.badge}
              </span>
            )}
          </Link>
        );
      })}
    </motion.nav>
  );
};
```

Description: This snippet shows the complex navigation logic for city officials. It handles the "Collapsible" sidebar state using Framer Motion for animations and integrates a Live Notification Badge system that calculates unread complaint counts in real-time to alert officials of new issues.

(v) Polymorphic Evidence Review Gallery

```

// Smart Evidence Detection and Dynamic Rendering
const renderEvidenceGrid = () => {
  return evidence.map((url, i) => {
    // Identifying file types via extension analysis
    const isAudio = url.toLowerCase().endsWith('.webm');
    const isVideo = url.toLowerCase().endsWith('.mp4') ||
url.toLowerCase().endsWith('.mov');

    return (
      <div key={i} className="evidence-card group overflow-hidden rounded-3xl border
border-slate-100 shadow-sm">
        {isAudio ? (
          /* Custom Voice Note Player with Official Styling */
          <div className="p-4 bg-blue-50/50 flex flex-col gap-2">
            <div className="flex items-center gap-2 text-blue-600">
              <Mic size={14} />
              <span className="text-[10px] font-black uppercase tracking-widest">Official
Voice Note</span>
            </div>
            <audio controls className="w-full h-8 accent-blue-600">
              <source src={resolveUrl(url)} type="audio/webm" />
            </audio>
          </div>
        ) : isVideo ? (
          /* Integrated Video Playback */
          <video controls className="w-full h-full object-cover">
            <source src={resolveUrl(url)} />
          </video>
        ) : (
          /* Image Gallery Preview with Hover Zoom */
          <img
            src={resolveUrl(url)}
            alt="Evidence"
            className="w-full h-full object-cover group-hover:scale-110 transition-
transform duration-500"
          />
        )}
      </div>
    );
  });
};

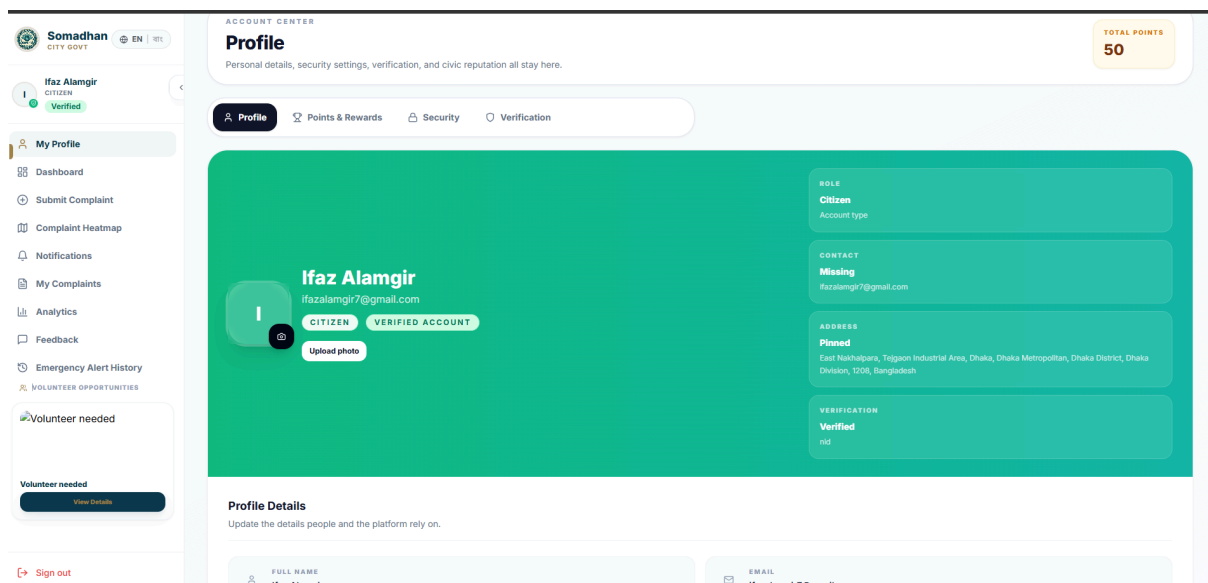
```

Description: This snippet demonstrates "Polymorphism" in the UI. Instead of just showing a list of links, the system intelligently detects the file type (Voice recording, Video, or Photo) and renders the appropriate interactive player. This is the crucial link that allows Public Servants to "hear" the citizen's voice note.

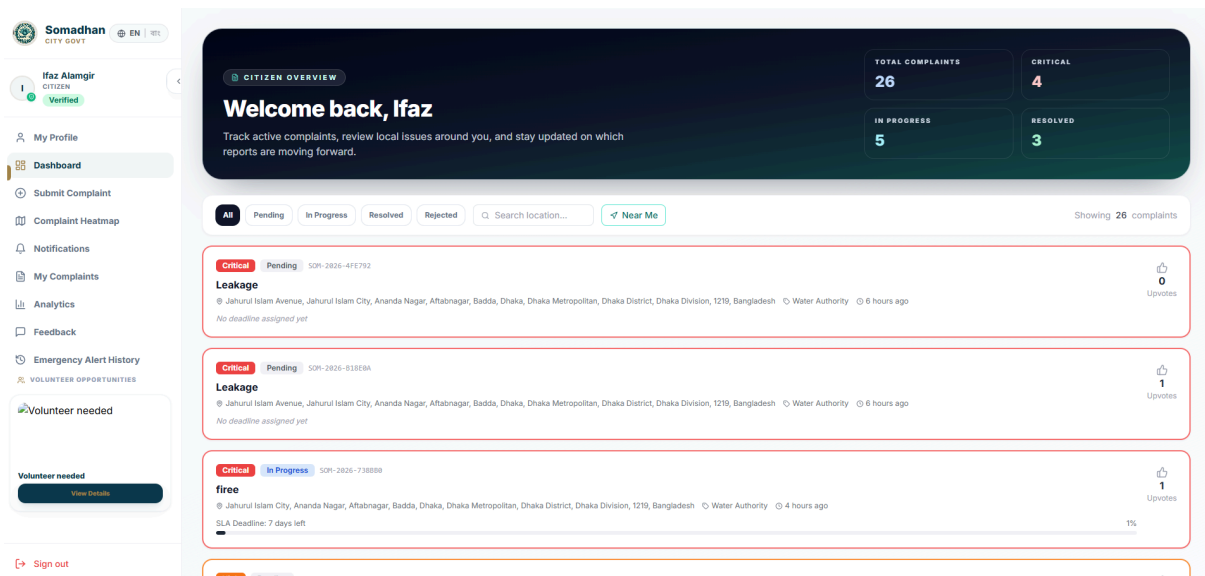
User Manual

Citizen User Manual

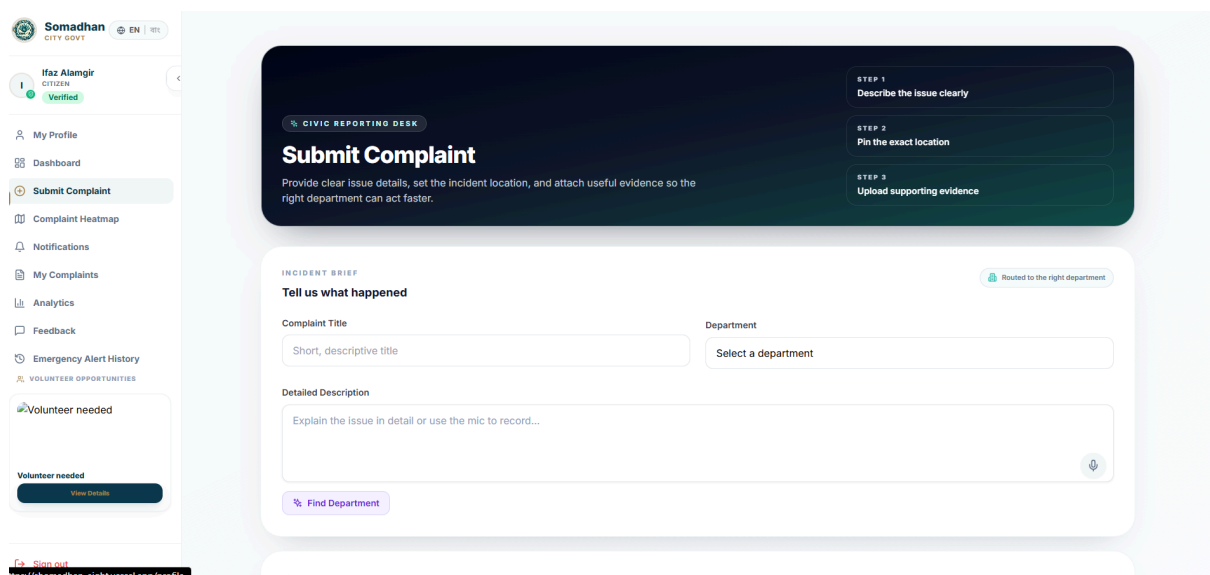
Navigate to the **My Profile** section to manage your personal identity, view your account verification status, and check your current civic reputation points. This page serves as your central hub for updating contact details and managing security settings to ensure your account remains secure and active.



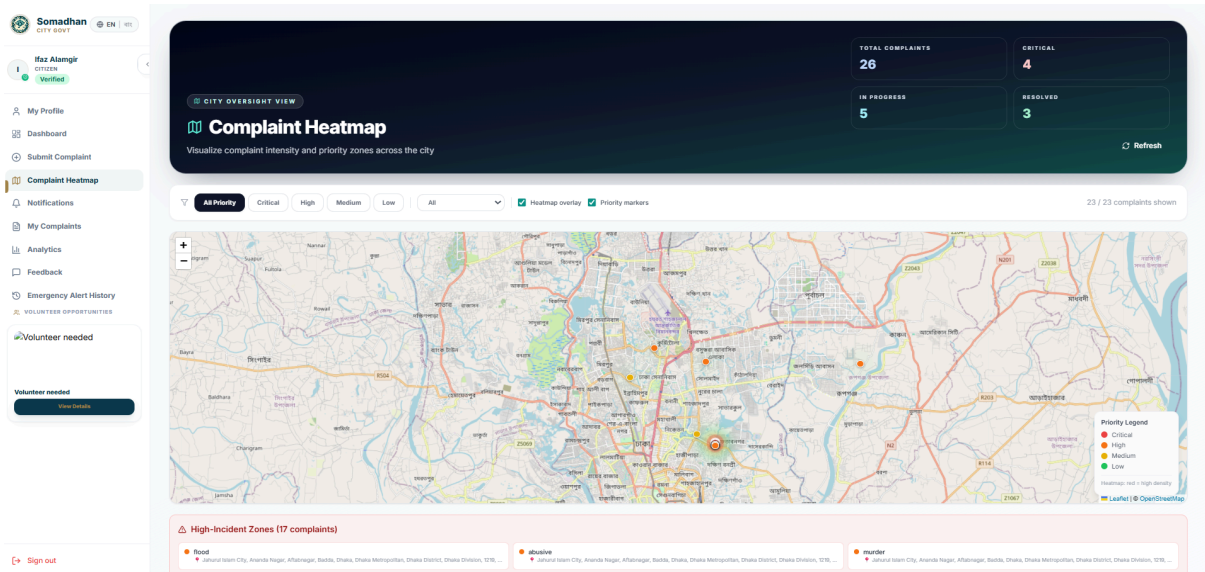
Use the **Dashboard** to get a quick overview of your activity, including the total number of complaints you have submitted and their current progress status. You can browse local issues, filter reports by status (such as Pending or Resolved), and use the "Near Me" feature to find incidents reported in your immediate area.



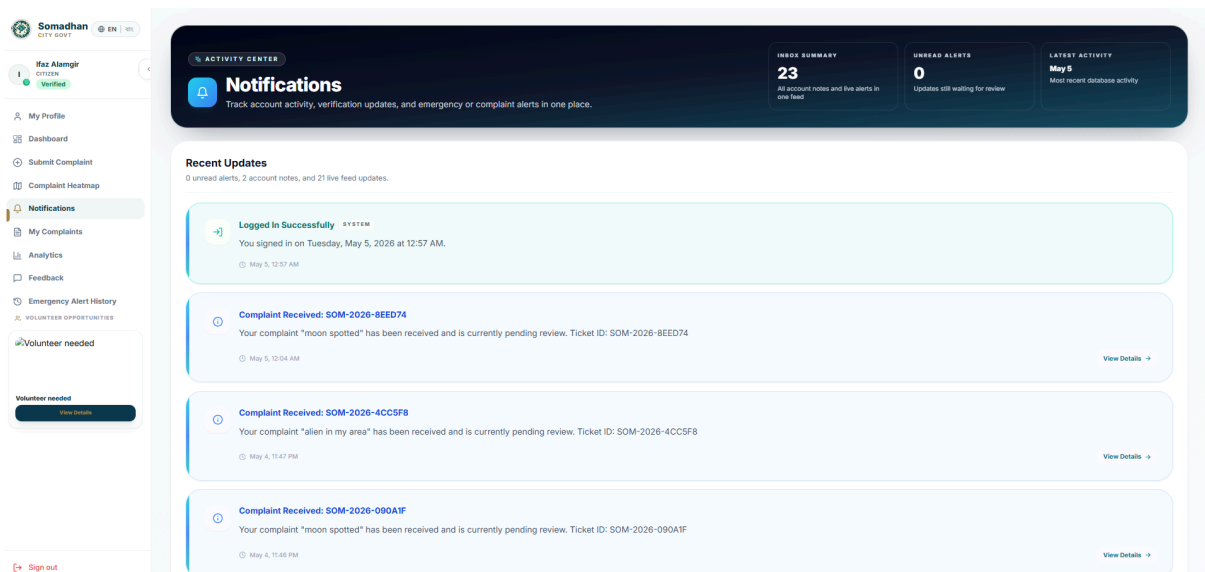
To report a civic issue, navigate to the **Submit Complaint** section and follow the three-step process: describe the issue clearly, pin the exact location, and upload supporting evidence. Fill in the descriptive title and select the appropriate department, or use the "Find Department" tool to automatically route your detailed description to the correct authorities.



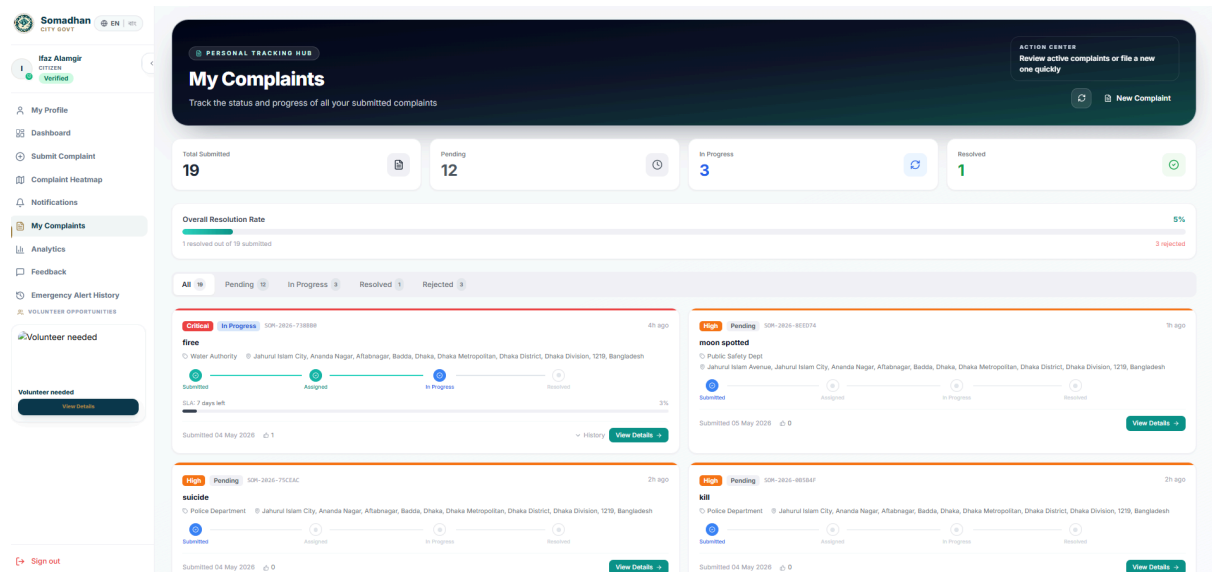
Access the **Complaint Heatmap** to visualize the intensity and location of civic issues across the city using an interactive map interface. Use the filter bar to sort incidents by priority level (Critical to Low) and toggle the heatmap overlay or priority markers to identify high-incident zones effectively.



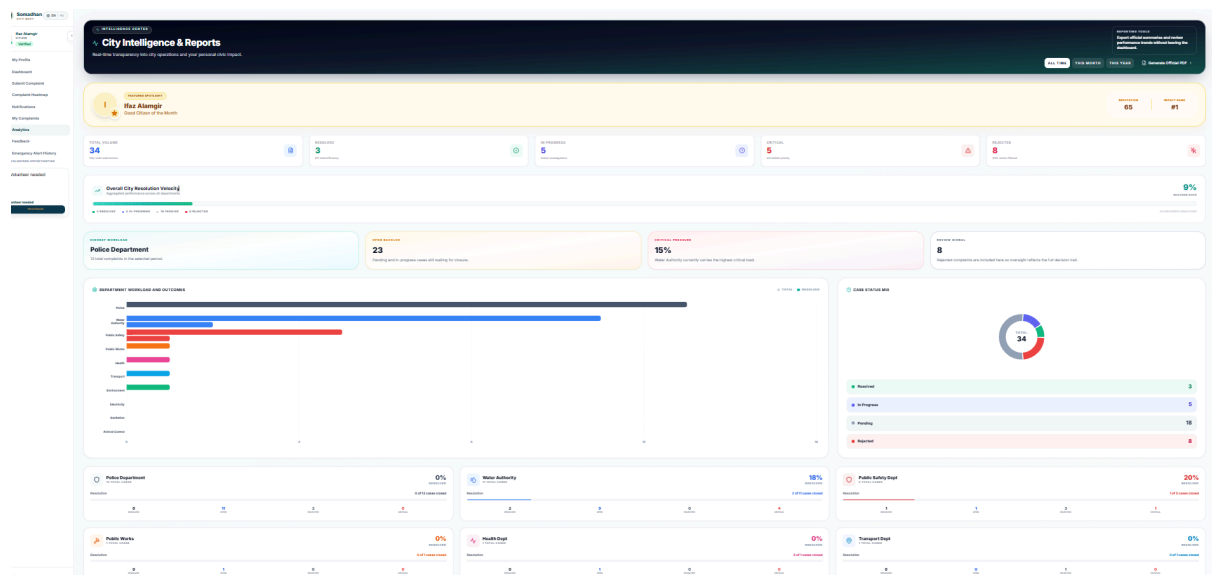
Visit the **Notifications** page to stay informed with real-time updates regarding your reported issues and city-wide announcements. You can easily sort through your alerts using the "All," "Updates," and "General" filters to track status changes, department assignments, and resolution confirmations.



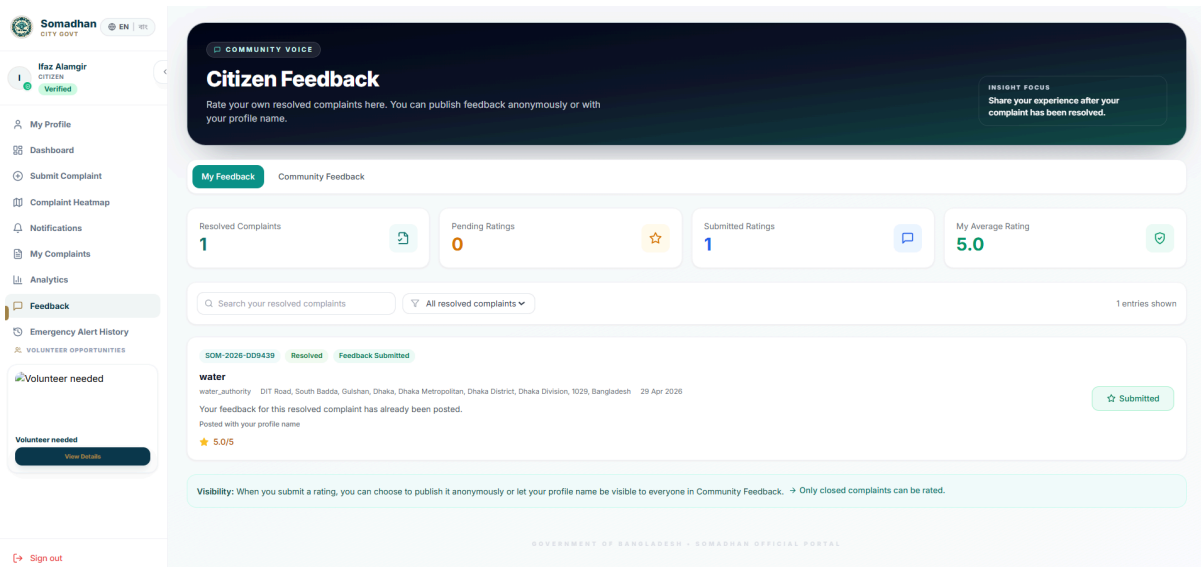
Access the **My Complaints** section to view a complete history of the issues you have reported, each identified by a unique ticket ID. Use the status tabs to monitor the lifecycle of your reports from "Pending" to "Resolved," and click on individual cards to see detailed timestamps and progress updates.



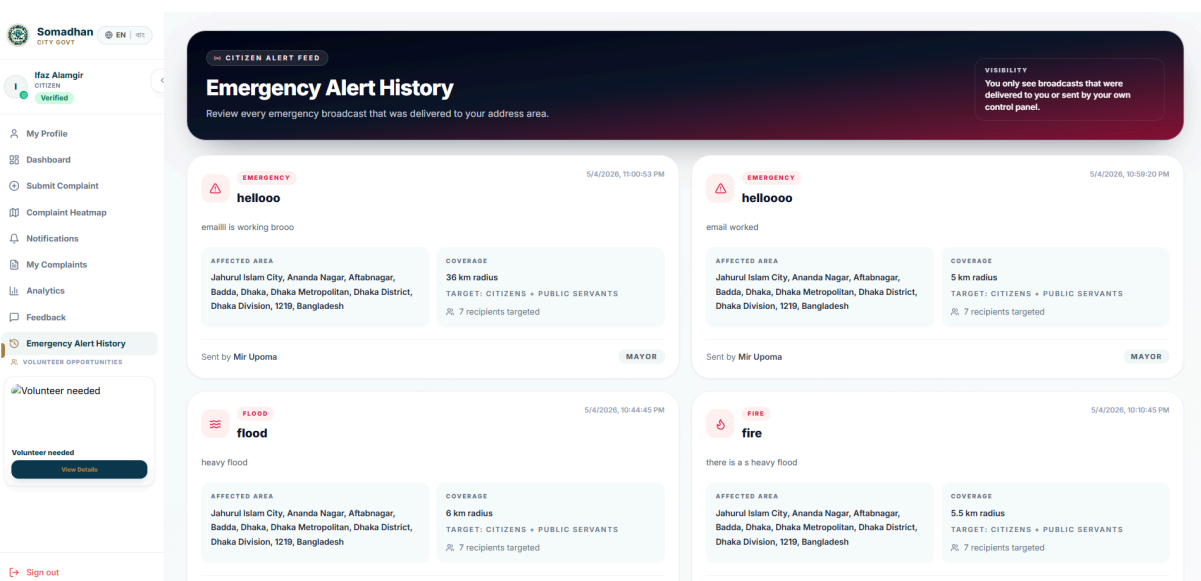
Navigate to the **Analytics** page to view a visual summary of civic performance, including a breakdown of complaint statuses and a comparison of reports filed across different city departments. Use the interactive charts to track resolution trends and see how various issues, such as utility failures or waste management, are being handled over time.



Go to the **Feedback** section to share your experience regarding resolved complaints and rate the quality of service provided by the city departments. Your ratings on response time and professionalism help improve civic standards and contribute to your reputation points within the community.

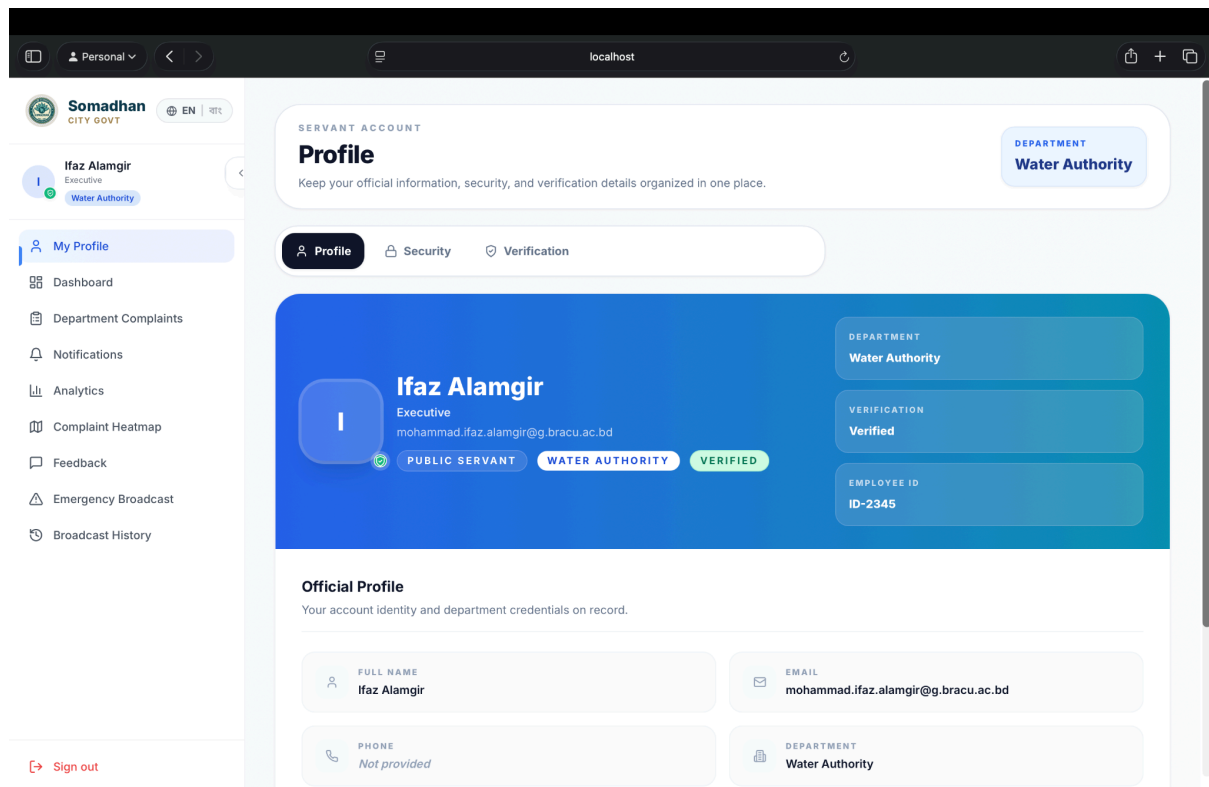


Explore the **Volunteer Opportunities** page to find civic engagement programs and community initiatives posted by city authorities. You can view detailed project descriptions, check registration requirements, and sign up to contribute your time toward local improvements and disaster response efforts.

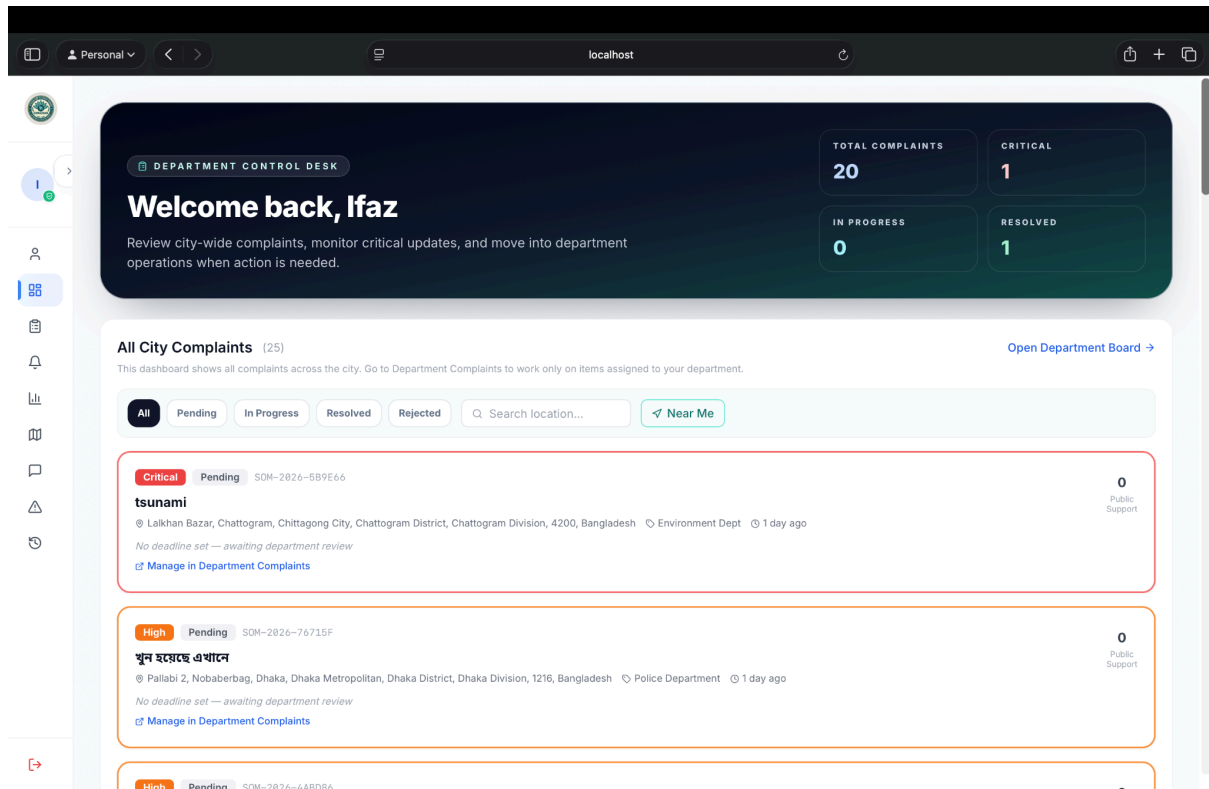


Public Servant User Manual

avigate to the **My Profile** section to manage your official servant account and verify your departmental credentials. This page displays your unique Employee ID, verified public servant status, and assigned department (e.g., Water Authority). Use the "Security" and "Verification" tabs to ensure your official information remains up-to-date and your account access is secure.



Use the **Department Control Desk** to monitor incoming civic reports and prioritize departmental actions. The dashboard provides a high-level summary of total complaints, critical incidents, and current resolution progress. Officers can browse "All City Complaints" to identify issues in their jurisdiction and select **Manage in Department Complaints** to assign a ticket to their specific operational queue for immediate review and field assignment.



Authorized officers must use the **Department Complaints** section to manage tasks specific to their assigned department. This interface provides a dedicated "Quick Overview" of your team's workload, including total cases and current resolution counts. Use the status filters to track pending reports, search by location, and use the **Generate Report** feature to export a comprehensive departmental summary for official documentation.

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Somadhan

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Ifaz Alamgir

Executive

Water Authority

My Profile

Dashboard

Department Complaints

Notifications

Analytics

Complaint Heatmap

Feedback

Emergency Broadcast

Broadcast History

Sign out

DEPARTMENT CONTROL CENTER

Department Complaints

5 complaints in your department

QUICK OVERVIEW

Total Cases	5
Pending	3
In Progress	0
Resolved	1

Generate Report

All Pending In Progress Resolved Rejected

Search by location...

Near Me

Medium Pending SOM-2026-46FA3D

হ্যালো

Sheikher Jayga, Trimohani, Nasirabad, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1214, Bangladesh

water_authority

1d ago

No resolution deadline set yet

0 UPVOTES

Medium Pending SOM-2026-C11C16

এটা হতে পারে সবচেয়ে ভয়ংকর রাত

Mirpur 7, Pallabi, Dhaka, Dhaka Metropolitan, Dhaka District, Dhaka Division, 1216, Bangladesh

water_authority

1d ago

No resolution deadline set yet

0 UPVOTES

Medium Pending SOM-2026-C22F86

হ্যালো কেমন আছেন ভূমি

American City, Kanchon, Rugganj Upazila, Narayanganj District, Dhaka Division, 1720, Bangladesh

water_authority

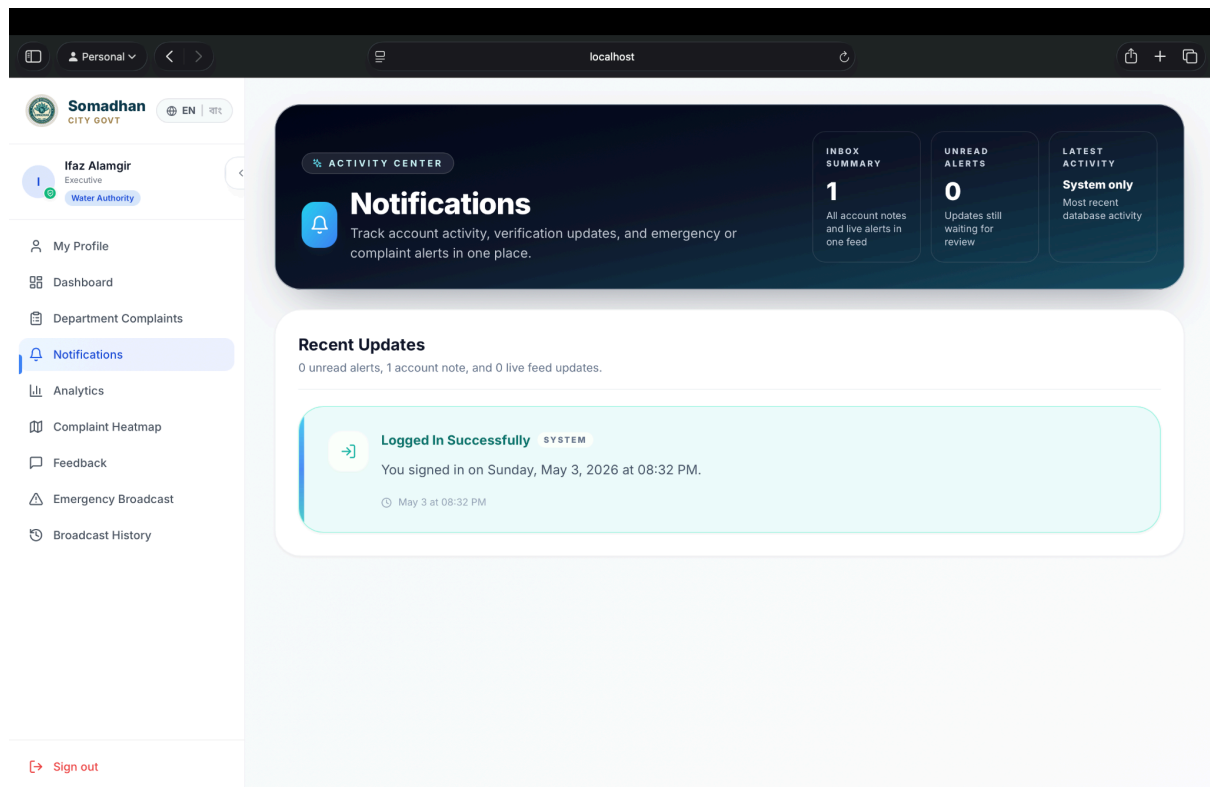
1d ago

No resolution deadline set yet

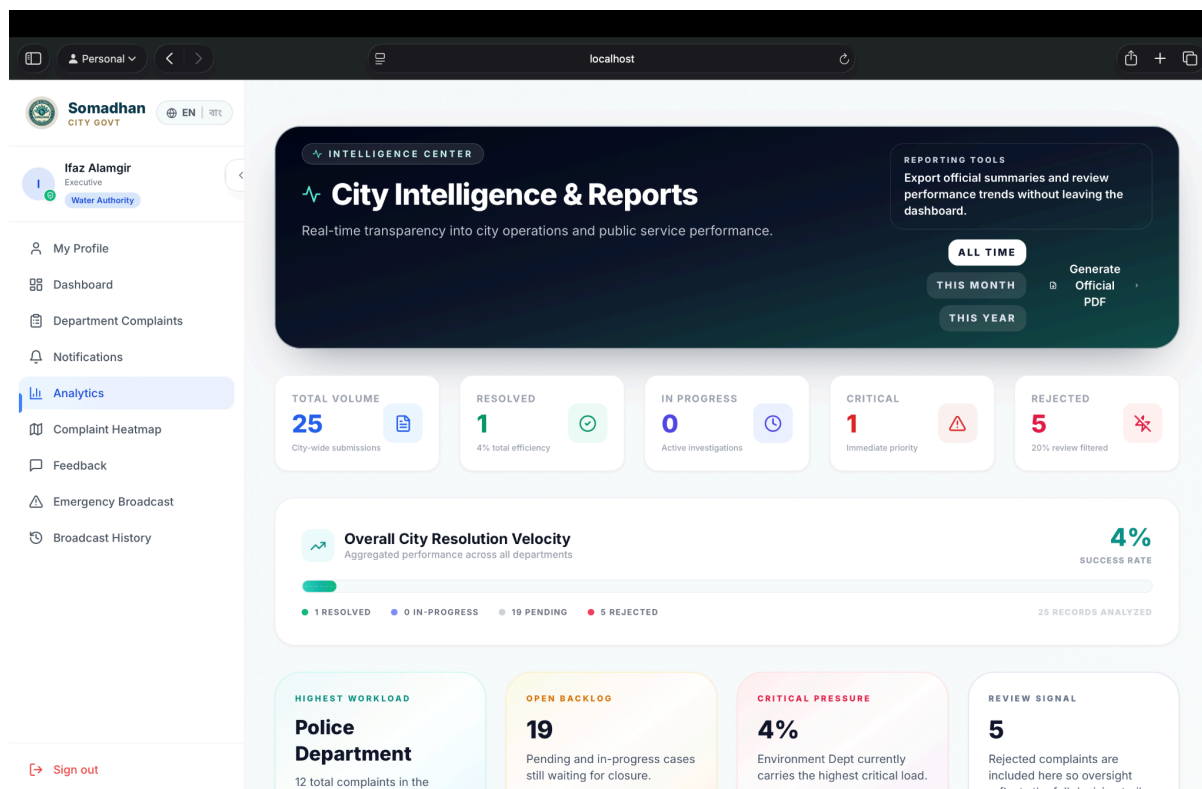
0 UPVOTES

Public servants should regularly check the **Notifications** section to stay informed about account activity, system updates, and urgent alerts. This hub provides an "Inbox Summary" of all live feed updates and unread alerts assigned to your profile. Use this page to track your login history and ensure you are aware of any new verification updates or high-priority complaint alerts that require departmental attention.

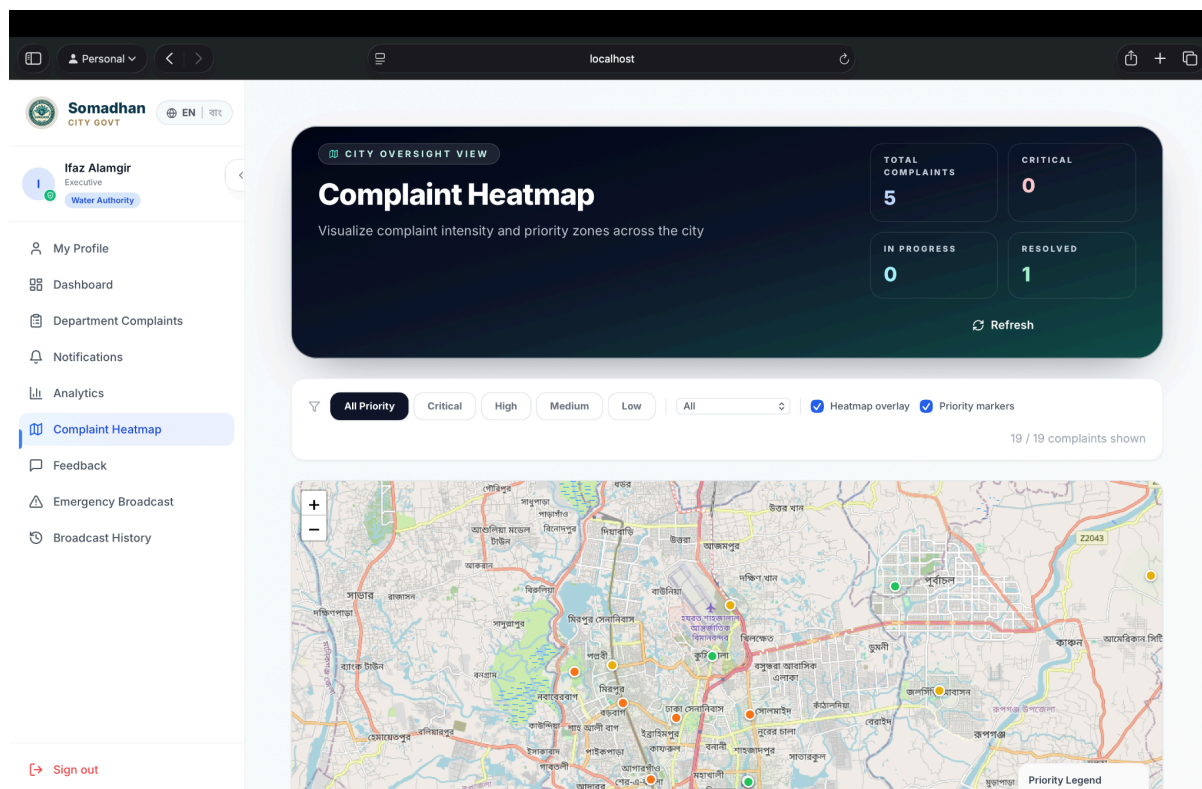
33



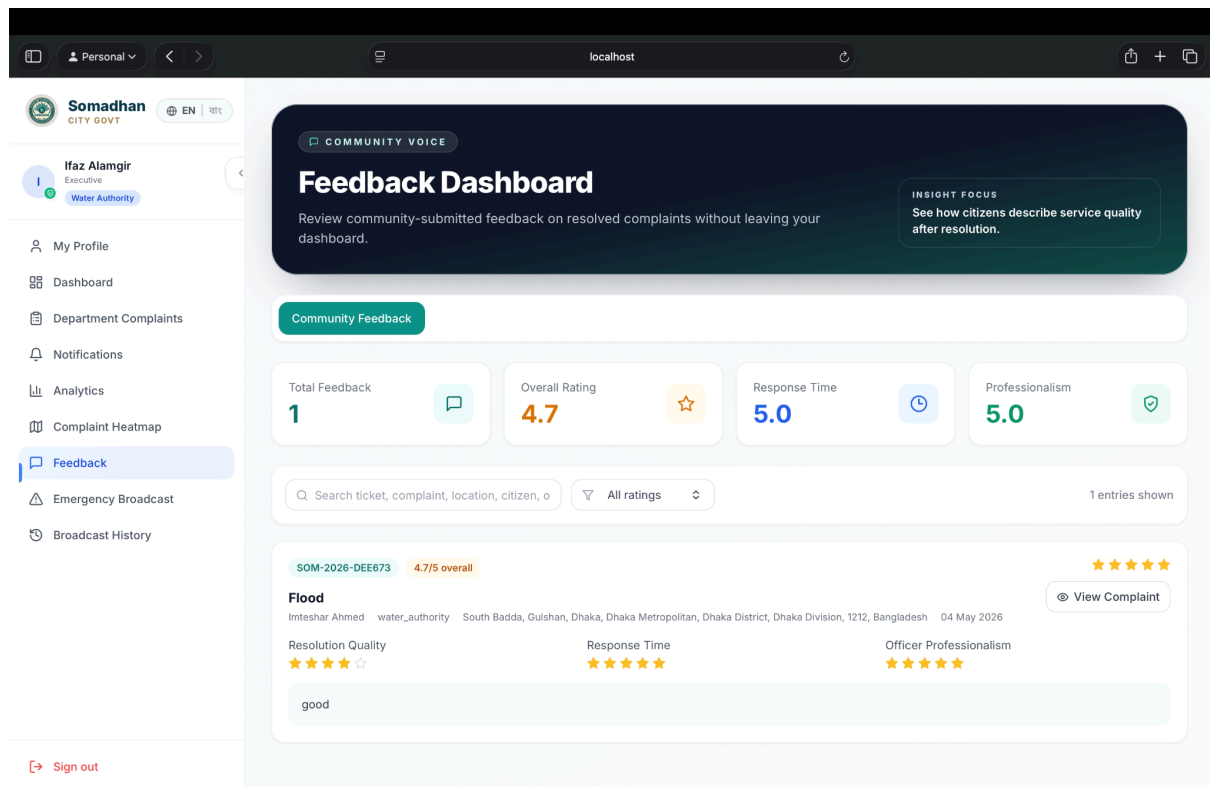
Use the **Analytics** section to monitor operational transparency and public service performance across all city departments. This dashboard provides a real-time summary of total complaint volumes, resolution efficiency, and critical-priority cases. Officers can filter data by specific timeframes (e.g., "This Month" or "This Year") and use the **Generate Official PDF** tool to export detailed performance trends and success rate analytics for administrative reporting and department audits.



Public servants can utilize the **Complaint Heatmap** to visualize complaint intensity and identify priority zones across the city. This geospatial tool allows you to filter reports by priority levels (Critical, High, Medium, Low) and toggle "Heatmap overlays" or "Priority markers" for a clearer view of problem areas. By monitoring these geographic clusters, departments can optimize resource allocation and strategically deploy field teams to high-density or high-priority incident locations.



Public servants can access the **Feedback Dashboard** to review community-submitted evaluations on resolved complaints. This section provides critical insight into service quality, displaying aggregate scores for **Response Time** and **Officer Professionalism**. Use this data to understand how citizens perceive your department's efficiency and to identify specific areas for operational improvement based on the ratings and comments provided for each ticket.



Authorized officers can access the **Emergency Broadcast** control panel to issue geo-targeted disaster alerts. To send an alert, provide a clear title, select the disaster type (e.g., Fire, Flood), and describe the immediate risk and necessary citizen actions in the message field. Use the "Affected Area" tool to pinpoint the emergency center on the map and adjust the **Live Radius** slider to ensure only citizens within the specified impact zone receive the notification.

Somadhan CITY GOVT EN | বাংলা

Ifaz Alamgir
Executive
Water Authority

- My Profile
- Dashboard
- Department Complaints
- Notifications
- Analytics
- Complaint Heatmap
- Feedback
- Emergency Broadcast**
- Broadcast History

[Sign out](#)

EMERGENCY BROADCAST CONTROL PANEL

Public Servant Emergency Desk

Send geo-targeted disaster alerts to citizens whose saved address falls inside the selected emergency radius.

TARGETING RULE
Only citizens with saved map coordinates inside the affected radius will receive the alert.

Create Emergency Broadcast

Fill in the incident details, choose the impact area, and send the alert.

ALERT TITLE
Flash flood warning near Dhanmondi

DISASTER TYPE
Fire

EMERGENCY MESSAGE
State what happened, the risk, and what citizens should do immediately.

AFFECTED AREA
Farazy Hospital Ltd, 15-19, Block-E, Rampura-Banasree Road, Block E, Banasree, Dhaka, Dhaka Metropolitan, Dhaka District, Dhaka D [Locate Area](#)

RADIUS (KM)
3

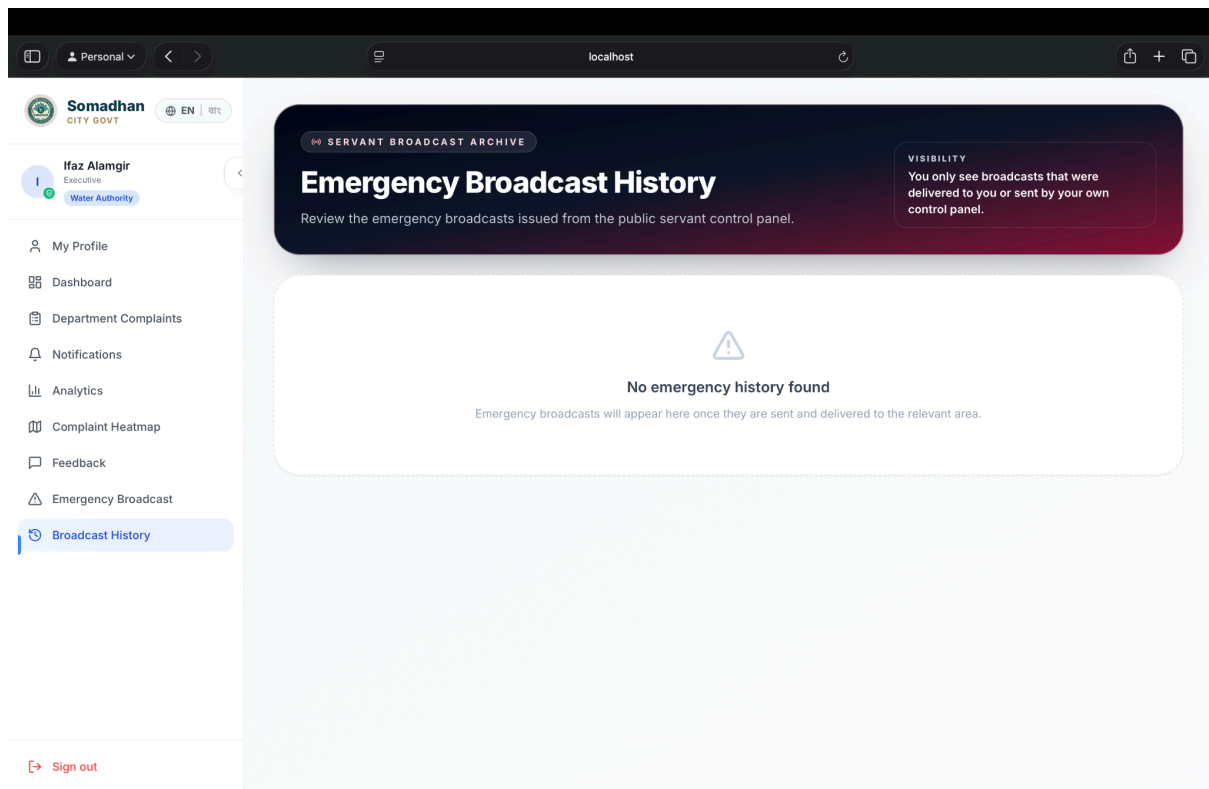
EXACT RADIUS
3

LIVE RADIUS
3.0 kilometers

Choose the emergency center point
Click anywhere on the map to update the target zone

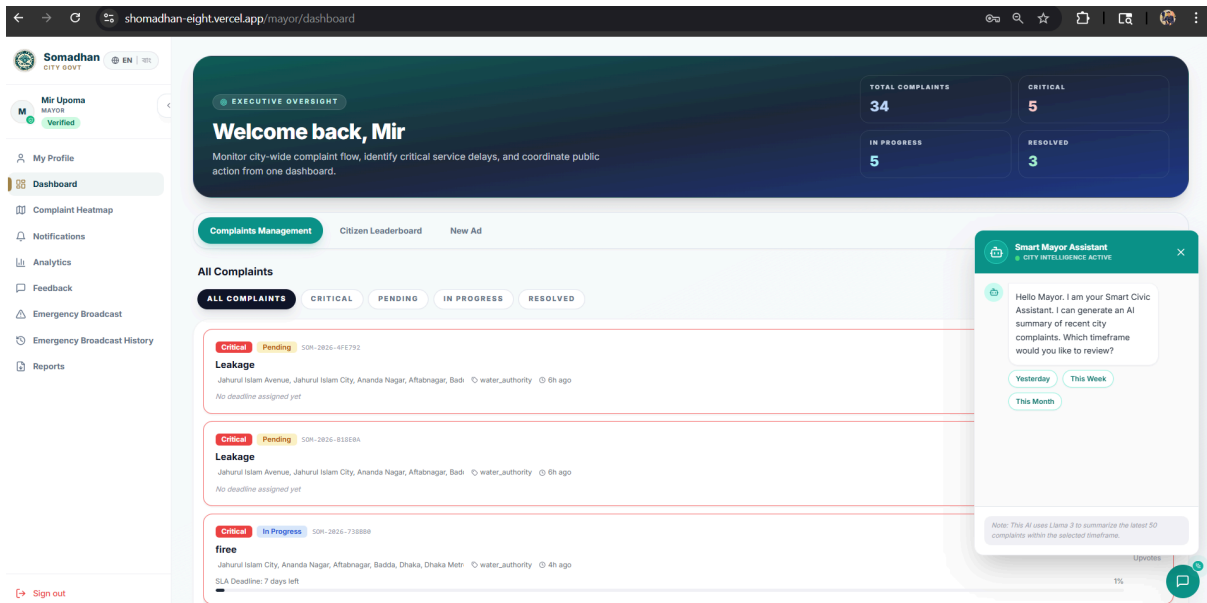
Emergency Broadcast History

The **Broadcast History** section serves as an official archive of all geo-targeted alerts issued by your department. This log provides visibility into past emergency communications, allowing you to review the timing, content, and delivery status of alerts sent to the public. This audit trail is crucial for departmental accountability and for analyzing the frequency and impact of emergency responses within your jurisdiction.

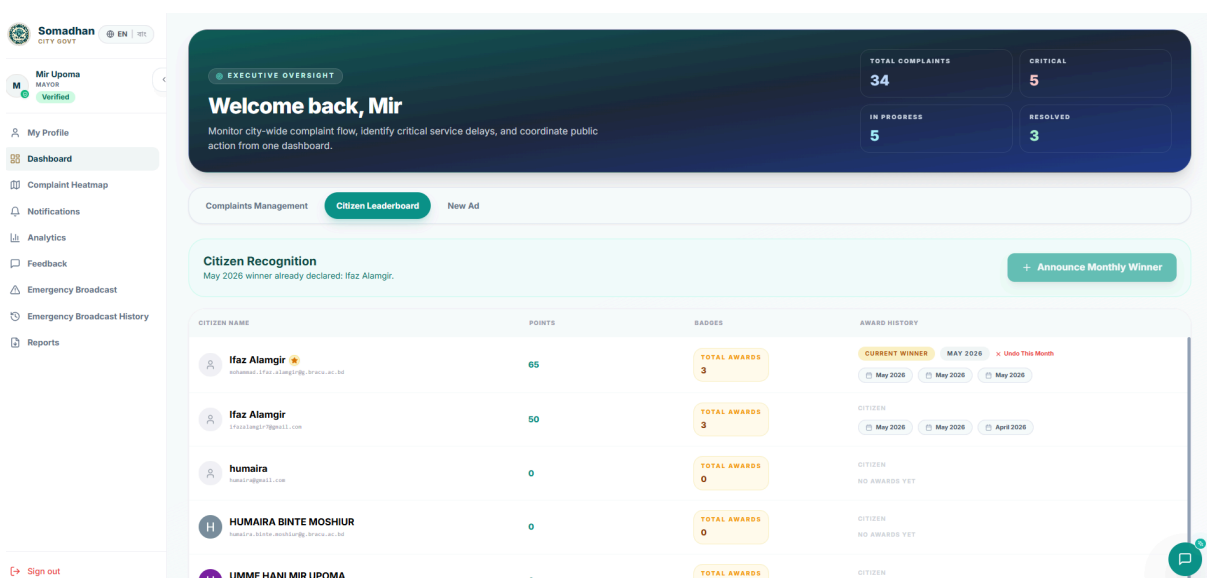


Mayor User Manual

The **Mayor's Dashboard** offers a centralized view of city-wide operations and departmental performance. Use the oversight panel to monitor real-time counts of **Critical** issues and overall resolution progress. Key features include the **Complaints Management** list for tracking specific incidents and the **Smart Mayor Assistant**, an AI tool that provides instant summaries of city activity over your preferred timeframe.



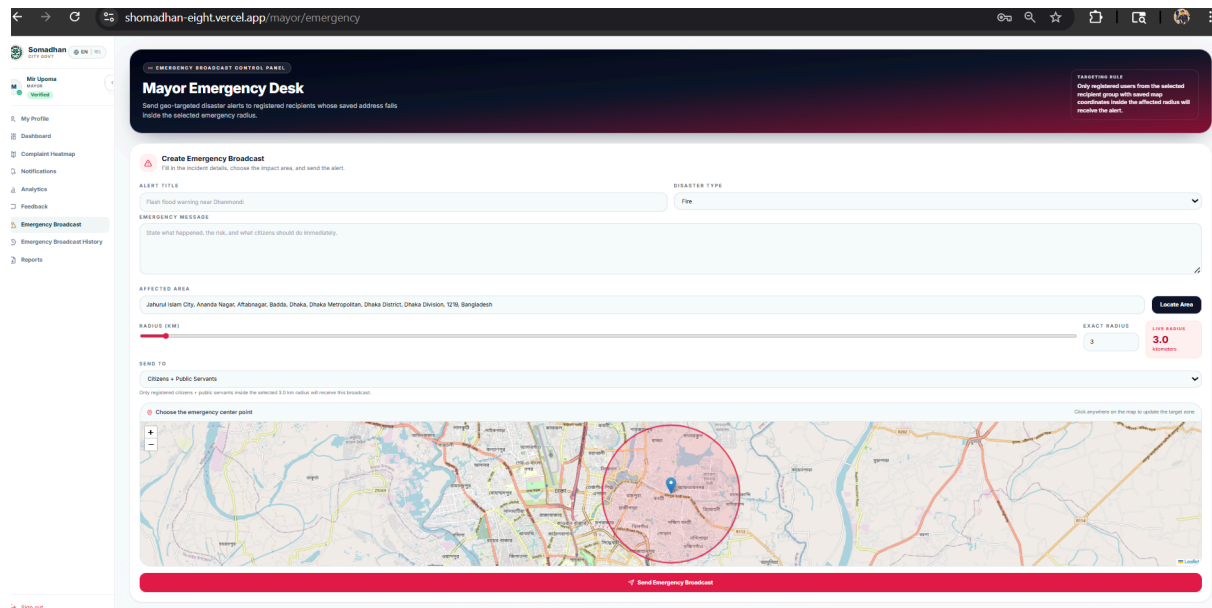
The **Citizen Leaderboard** allows the Mayor to track community engagement and reward top contributors. From this view, you can see a list of citizens ranked by their earned points and badges. Use the **Announce Monthly Winner** button to officially recognize the most active participant, fostering a culture of civic responsibility and rewarding those who consistently help improve the city.



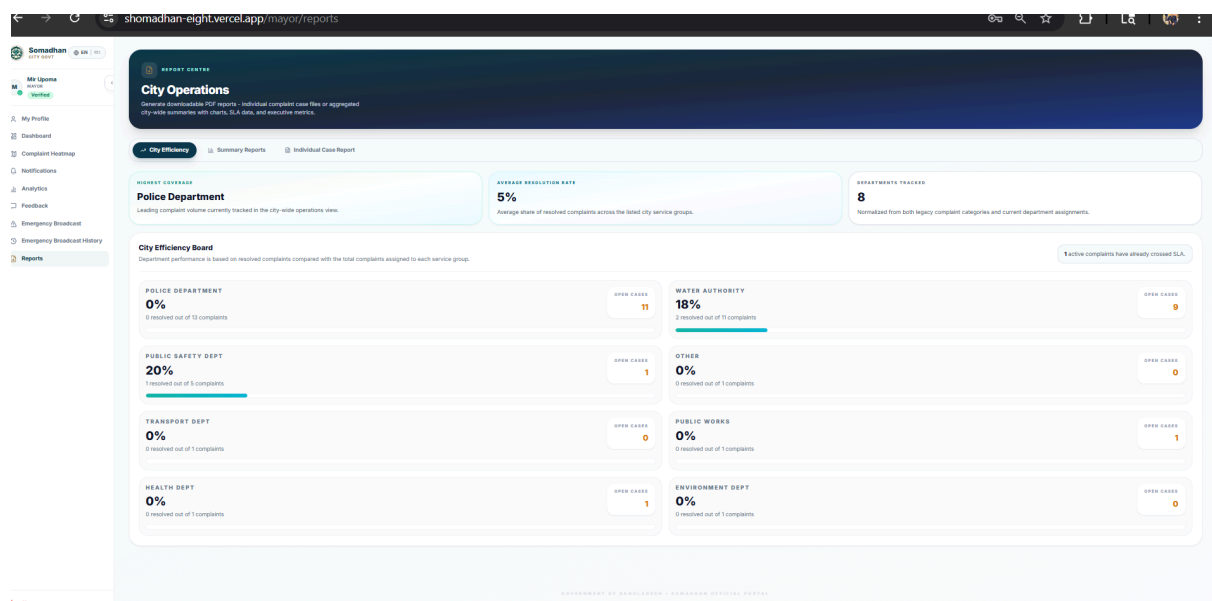
The **New Ad** tab allows the Mayor to publish official city volunteer advertisements for initiatives like flood response or public cleanup drives. To create a campaign, fill in the **Title**, **Description**, and **Event Date**, and specify the required **Headcount**. You can also upload an official poster and provide contact details before clicking **Publish** to share the announcement with all citizens.

The screenshot displays the 'Somadhan CITY GOVT' dashboard for Mayor Mir Upoma. The left sidebar lists navigation options: My Profile, Dashboard, Complaint Heatmap, Notifications, Analytics, Feedback, Emergency Broadcast, Emergency Broadcast History, and Reports. The main content area features a 'Welcome back, Mir' message and a 'EXECUTIVE OVERSIGHT' section with statistics: 34 Total Complaints, 5 In Progress, 5 Critical, and 3 Resolved. Below this is a 'New Ad' button. The 'New Ad' form is titled 'Publish a new city volunteer advertisement' and includes instructions, suggested use (flood response, public cleanup drives, health camps, and neighborhood outreach), and best practice (use a clear event title, exact date, and a reachable phone or email before publishing). The form fields are: Title, Description, Event Date (mm/dd/yyyy), Headcount (10), Contact, and Upload Poster. A 'Publish' button is at the bottom right.

The **Mayor Emergency Desk** allows for the issuance of geo-targeted alerts to citizens and public servants during disasters. To deploy an alert, enter a title, select the disaster type, and provide clear instructions in the emergency message. Use the map to pinpoint the **Affected Area** and adjust the radius slider to define the exact impact zone; once set, click **Send Emergency Broadcast** to notify all registered users within that specific range.

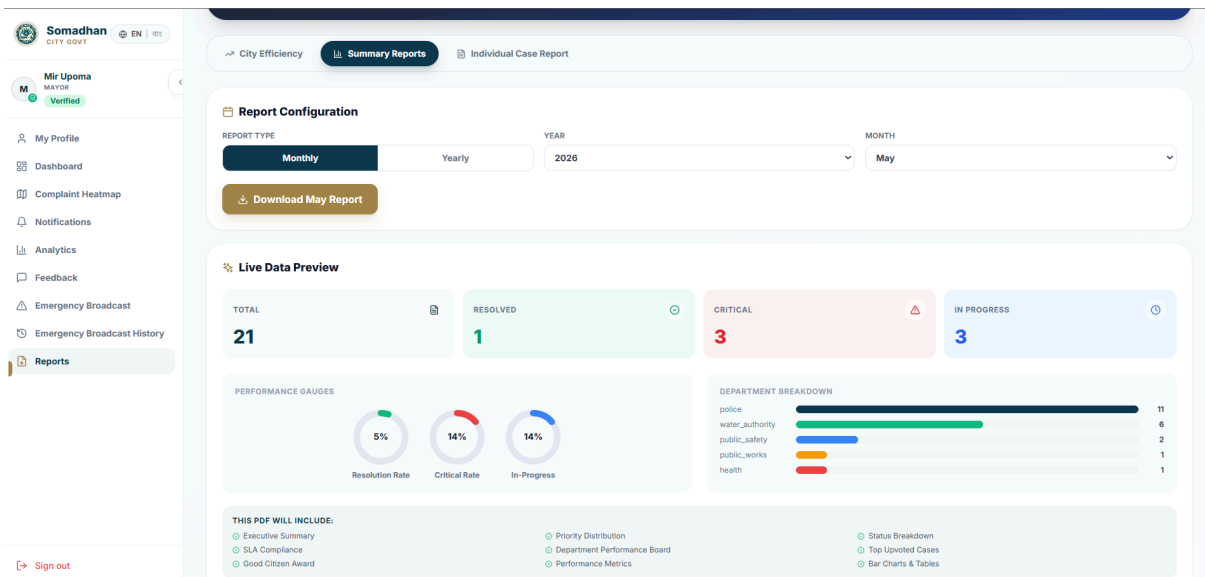
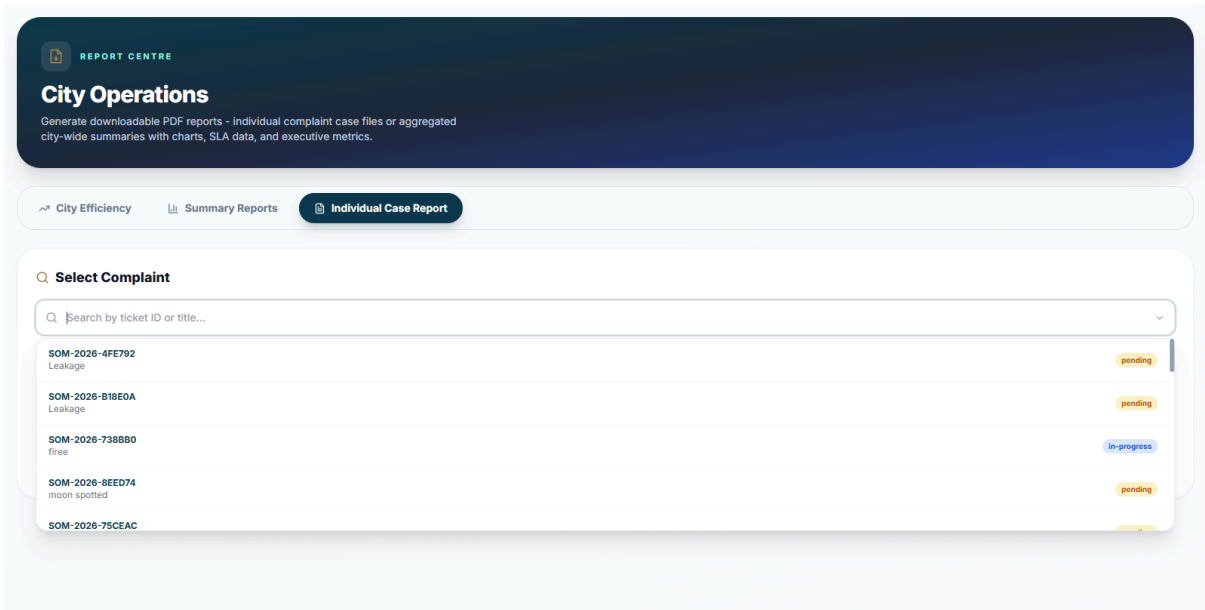


The **Reports** section provides the Mayor with a comprehensive overview of departmental efficiency and city-wide performance. Use the **City Efficiency Board** to track resolution rates across different sectors, such as the Police Department or Water Authority. From this panel, you can identify which departments have the highest workloads or pending cases and generate downloadable PDF reports—including **Summary Reports** and **Individual Case Reports**—to assist in executive decision-making and performance reviews.



Within the **Report Centre**, the Mayor can access the **Individual Case Report** feature to drill down into specific civic issues. This tool allows you to search for a complaint by its unique **Ticket ID** or title (e.g., "Leakage" or "fire"). By selecting an entry from the list, you can view

the full history and current status—such as **pending** or **in-progress**—of any individual report to ensure precise oversight and accountability for specific city incidents.

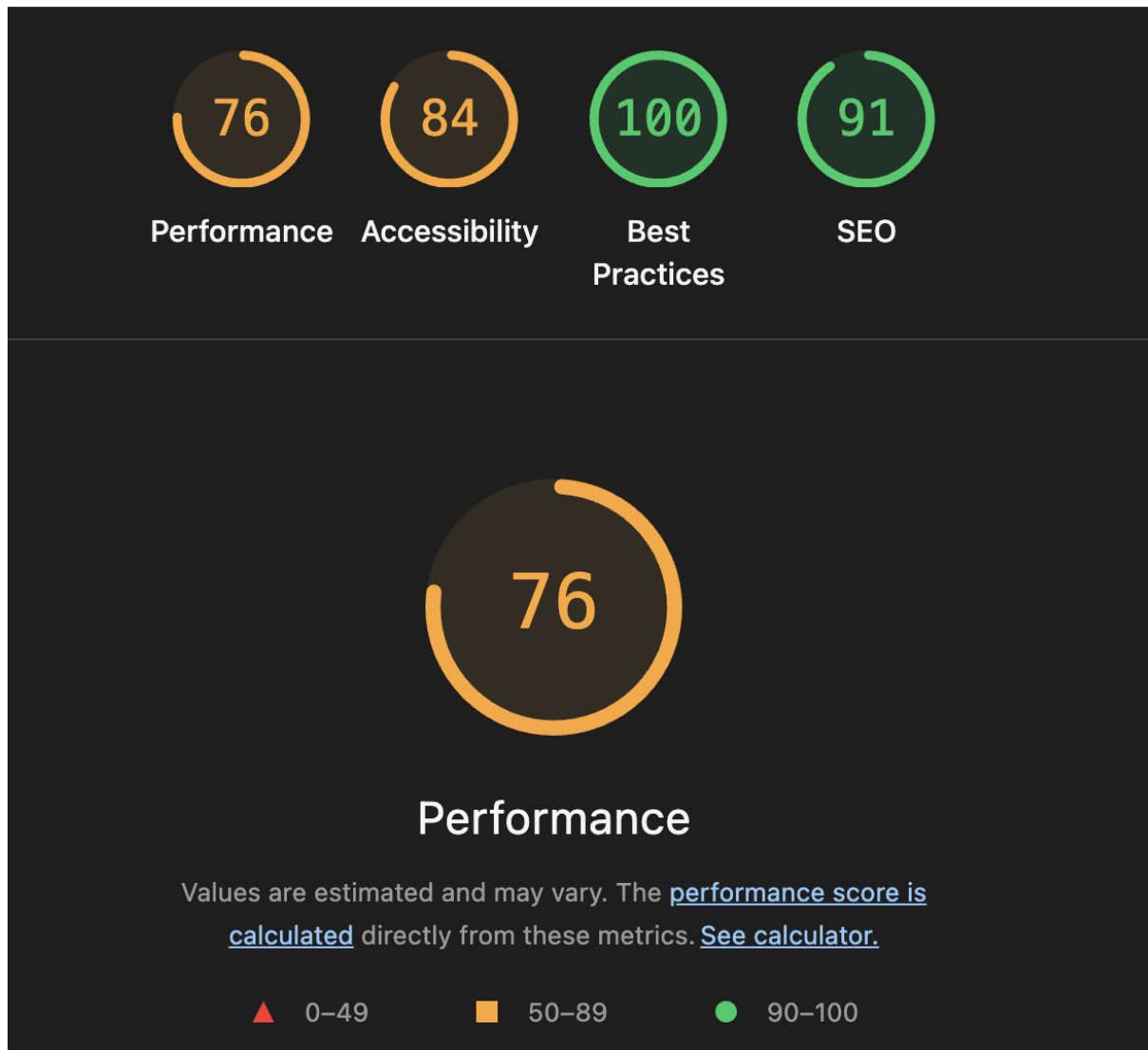


The **Summary Reports** tab allows the Mayor to generate high-level analytical documents for executive review. Within the **Report Configuration** panel, you can select between "Monthly" or "Yearly" report types and specify the exact year and month required. Once configured, use the **Download Report** button to export a comprehensive PDF.

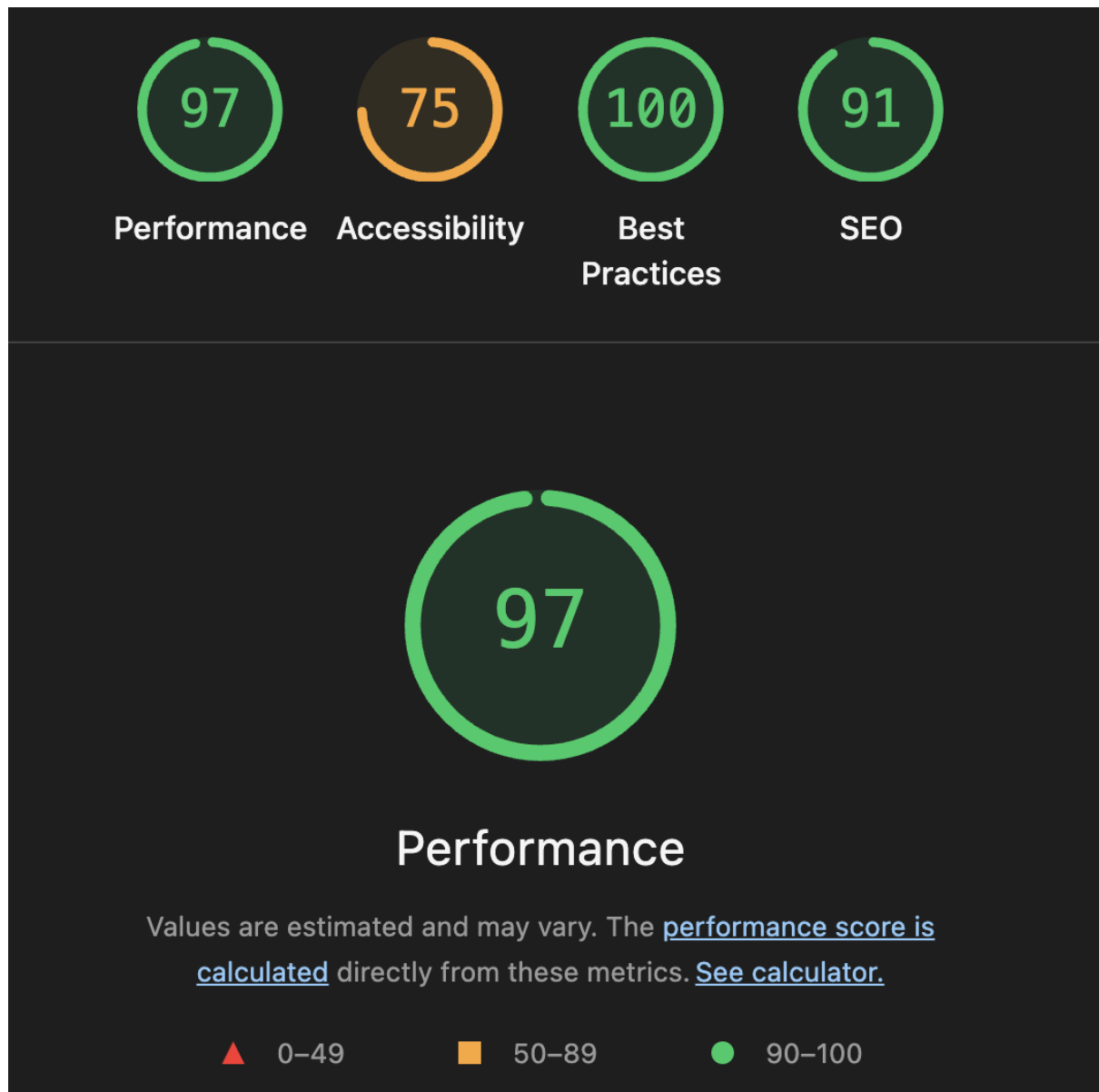
Performance and Network Analysis

Generate Performance and Network analysis report using lighthouse DevTool. Add screenshots of the report. Also, attach a screenshot of your system in any mobile viewport. **The system UI must be responsive.**

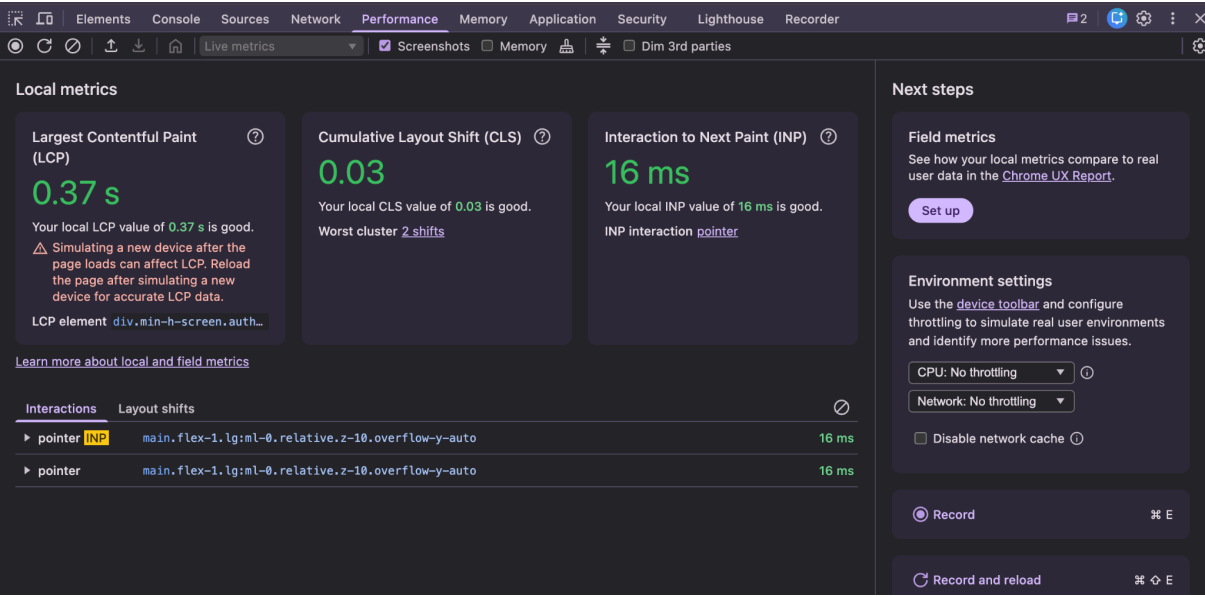
(Mobile Analysis):



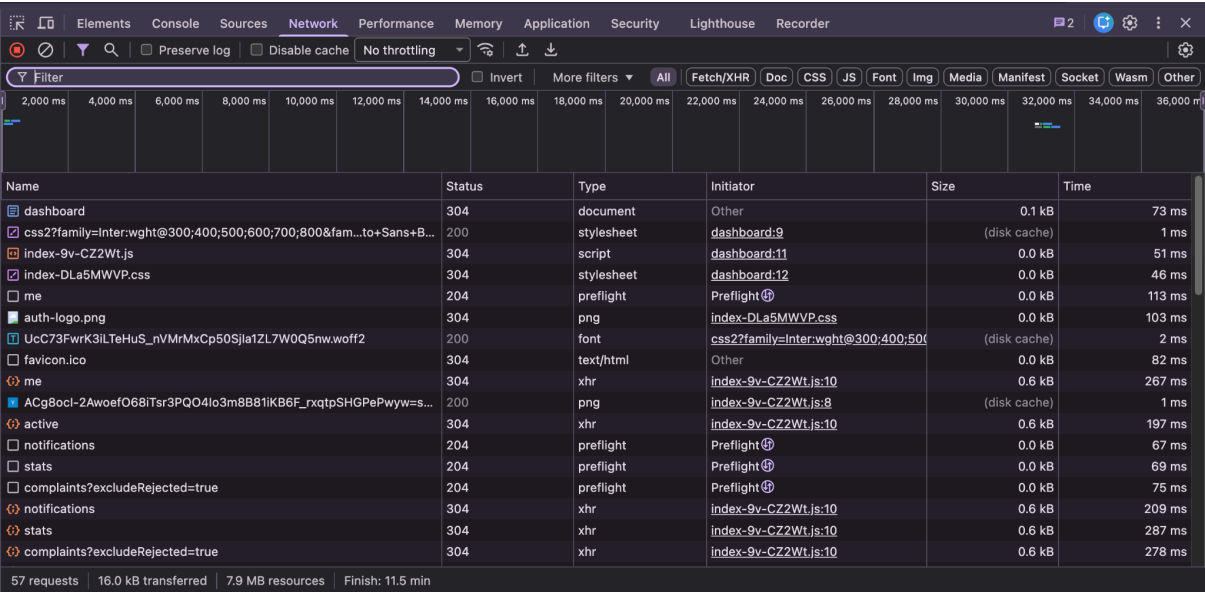
(Desktop Analysis):



Performance Metrics:



Network Metrics:



Github Repo [Public] Link

<https://github.com/ifaz07/Shomadhan>

Link of Deployed Project

<https://shomadhan-eight.vercel.app/>

Individual Contribution

Group member - 01	
Name: Umme Hani Mir Upoma	Student ID:22241025
Functional Requirements which are developed by this member:	
1.The system shall support multilingual interfaces (Bangla and English) with real-time translation for complaint descriptions using an external translation API to ensure accessibility for diverse users.	
2.The system shall automatically determine complaint priority levels (Low, Medium, High, or Critical) based on severity, location sensitivity, category type, and emergency flags, and also visualize complaint intensity using a heatmap overlay on Google Maps API to identify high-incident zones within the city(Public Voting Mechanism-The system shall allow citizens to upvote public complaints to increase their visibility and influence priority scoring.	
3. Emergency Broadcast Control Panel -->The system shall allow authorities to send geo-targeted push notifications during disasters (fire, flood, road collapse)	
4.The system shall provide an integrated dashboard allowing citizens access aggregated public analytics including total complaints, resolved and pending cases, and department-wise performance statistics.	

Group member - 02	
Name: Mohammad Ifaz Alamgir	Student ID: 22241022
Functional Requirements which are developed by this member:	

1. Duplicate Complaint Detection and Spam Filtering: The system shall implement duplicate complaint detection and spam filtering mechanisms by analyzing similar keywords, textual content, and geographic proximity of submitted complaints. This functionality shall prevent redundant ticket generation and ensure data integrity within the system.

2. The system shall display similar existing complaints within a predefined geographic radius before allowing a user to submit a new complaint. This feature shall encourage users to upvote or support existing issues instead of creating duplicate entries. Additionally, the system shall implement Service Level Agreement (SLA) monitoring by assigning resolution deadlines to complaints based on their respective category types. The system shall continuously track and display the remaining resolution time and notify relevant authorities when deadlines are approaching or exceeded.

3. The system shall provide an integrated citizen dashboard that enables users to view their complaint history, resolution details, and department assignment information. The system shall also provide a centralized authority dashboard (e.g., Police Department, Water Authority, Municipal Corporation) where assigned complaints can be viewed, updated, marked as resolved, and supplemented with field notes and supporting proof images.

4. The system shall provide an integrated dashboard that allows citizens to track the status of their complaints in real time. Additionally, the system shall implement a configurable workflow engine that supports predefined state transitions, including Pending, Assigned, In Progress, Resolved, and Closed. The system shall record and log each status transition along with corresponding timestamps and responsible personnel information to ensure transparency, traceability, and accountability throughout the complaint lifecycle

Group member - 03

Name: Imteshar Ahmed

Student ID: 22241016

Functional Requirements which are developed by this member:

1. Implemented an NLP-driven engine to automatically categorize complaints, extract key terms, and route tickets to the appropriate departments for faster processing.

2. Developed an automated validation system that filters prank submissions and escalates overdue tasks to higher authorities based on priority and age.

3. Built the quality of resolution rating system and a reputation-based gamification engine to encourage active and honest citizen participation (feedback).

4. Engineered the comprehensive PDF generation module, featuring specialized analytics and corruption-risk prediction based on ticket delay patterns.

Group member - 04

Name: Humaira Binte Moshir

Student ID: 22241022

Functional Requirements which are developed by this member:
1. The system shall allow citizens to submit complaints by providing detailed descriptions, selecting complaint categories, and uploading supporting multimedia evidence such as images, videos, or voice records (Anonymously) while capturing GPS-based location data using Google Maps API, and simultaneously generate a unique ticket ID, record submission. timestamp, assign an initial status of “Pending,” and store complete complaint metadata in the database
2. The system shall implement a centralized notification system that sends real-time alerts to citizens and officers via SMS, email, or in-app messages when tickets are created, updated, escalated, or resolved, and additionally broadcast area-based emergency notifications within a defined geographic radius for high-risk or emergency complaints, subject to category rules or authority approval.
3. Mayor / City Corporation Dashboard, a centralized monitoring panel for authorities to view all city complaints, department performance, police case progress, pending versus resolved issues, SLA compliance, and heatmap analysis of high-problem areas, along with a volunteer advertisement system where authorities can post civic announcements and citizens can register as volunteers.
4. The system shall provide a civic reward or reputation system where active and verified users earn credibility badges for responsible reporting and community participation.

References

N/A